

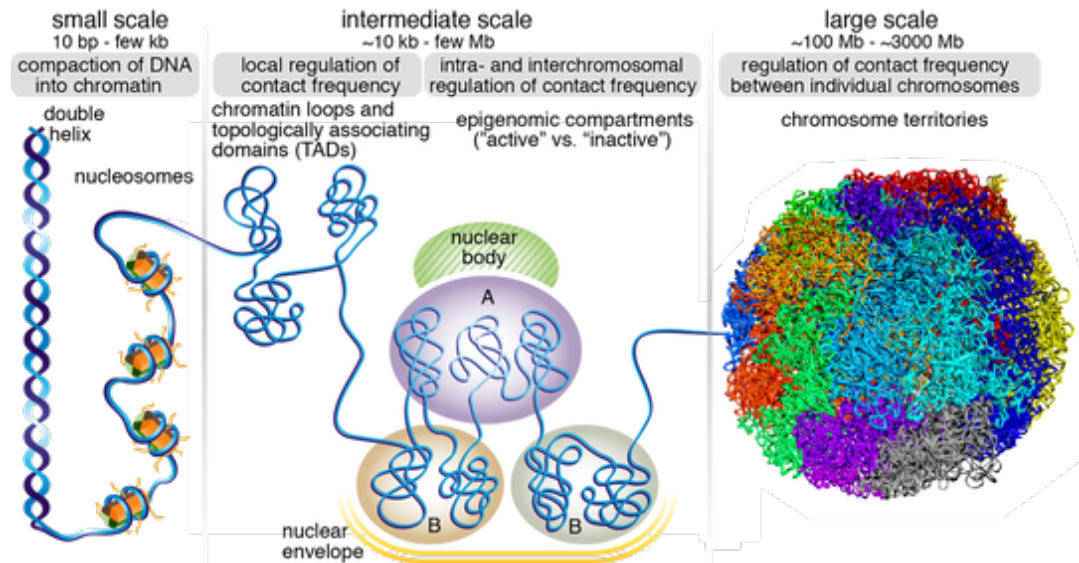
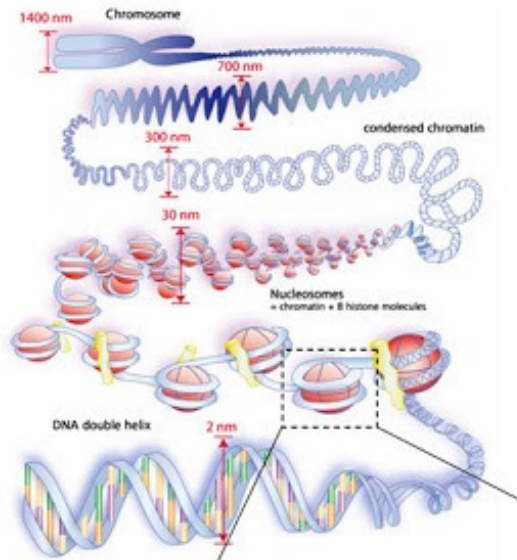


3rd Generation sequencing techniques

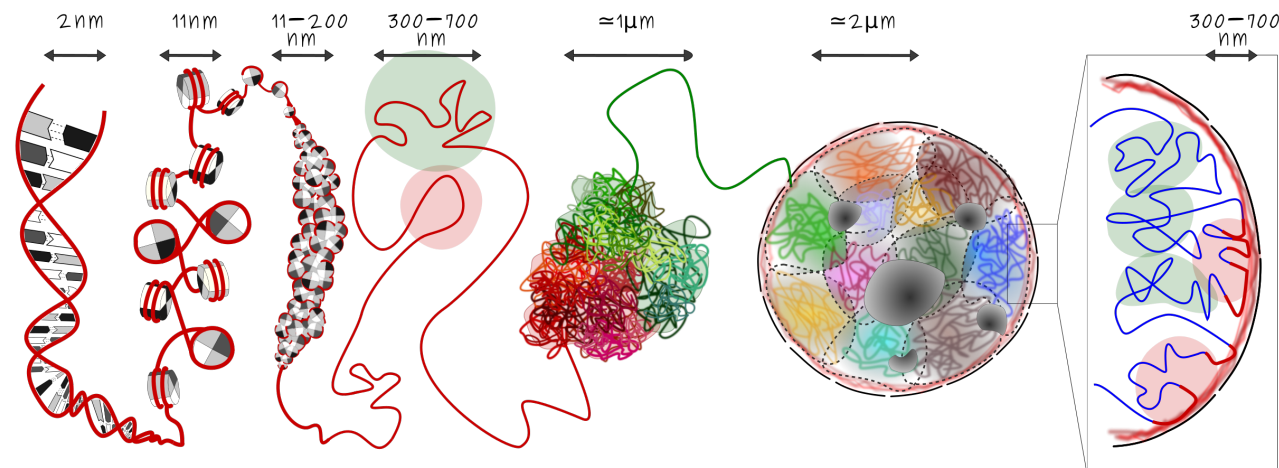
Chapter 4



Genomics



DNA Chromatin fiber DNA loop & Compartment, Chromatin/TAD domains Chromosome territory inside cell nucleus Lamina-associated domain (LAD)



- Nucleobases
- Nucleosome, composed of 8 histones protéines
- Repressive chromatin domain, mainly heterochromatin
- Active chromatin domain, mainly euchromatin
- Chromosome territory
- Nucleolus
- Nuclear speckles
- Nuclear membrane, including schematic nuclear pores
- Nuclear lamina
- LAD



2022 Spring Week 1-2



1. Introduction of Genomics

2. Introduction of the Human Genome Project (HGP)

3. Second-generation sequencing techniques (Homework 1)

4. Third-generation sequencing techniques

9. The annotation of chromatin states

10. The detecting techniques for chromatin states

11. The annotation and detection method for enhancer, promoter, and insulator (Homework 2)

12. The long-range interactions of 3D Genome

5. The structure of genome

6. The functional element of genome

7. The stage of the functional element annotation for genome-1

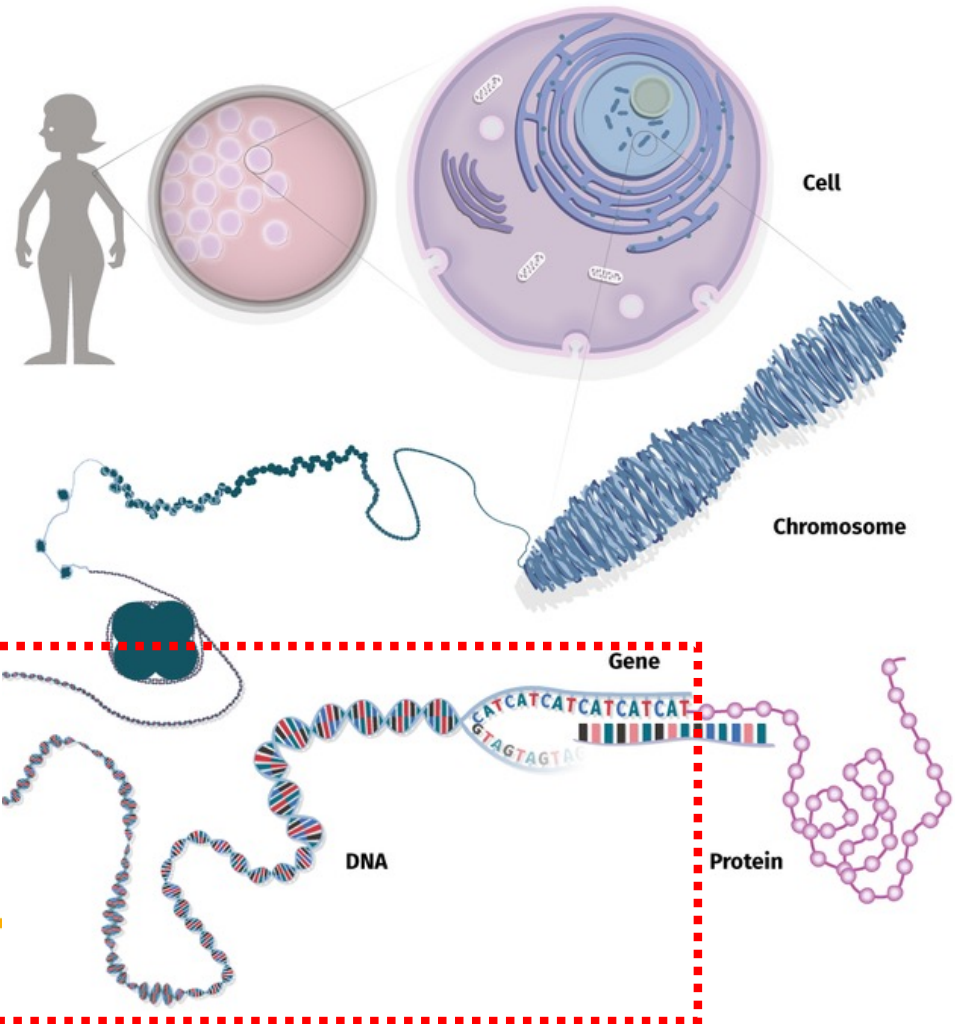
8. The stage of the functional element annotation for genome-2

13. The interaction techniques for 3D Genome

14. The interaction techniques for the specific factor mediated long-range interaction for 3D Genome (Homework 3)

15. The new generation of 3D Genome technology

16. The introduction of Genomics & epigenetics-1



21. The techniques of single-cell transcriptome

22. Genome and transcriptome (Homework 5)

23. Genomics and Cancer research

24. Write a review about the application and development of Genomics

Where we are?

2nd-gen sequencing method (go over)

- 454 pyrosequencing



- SOLID sequencing



- Illumina sequencing



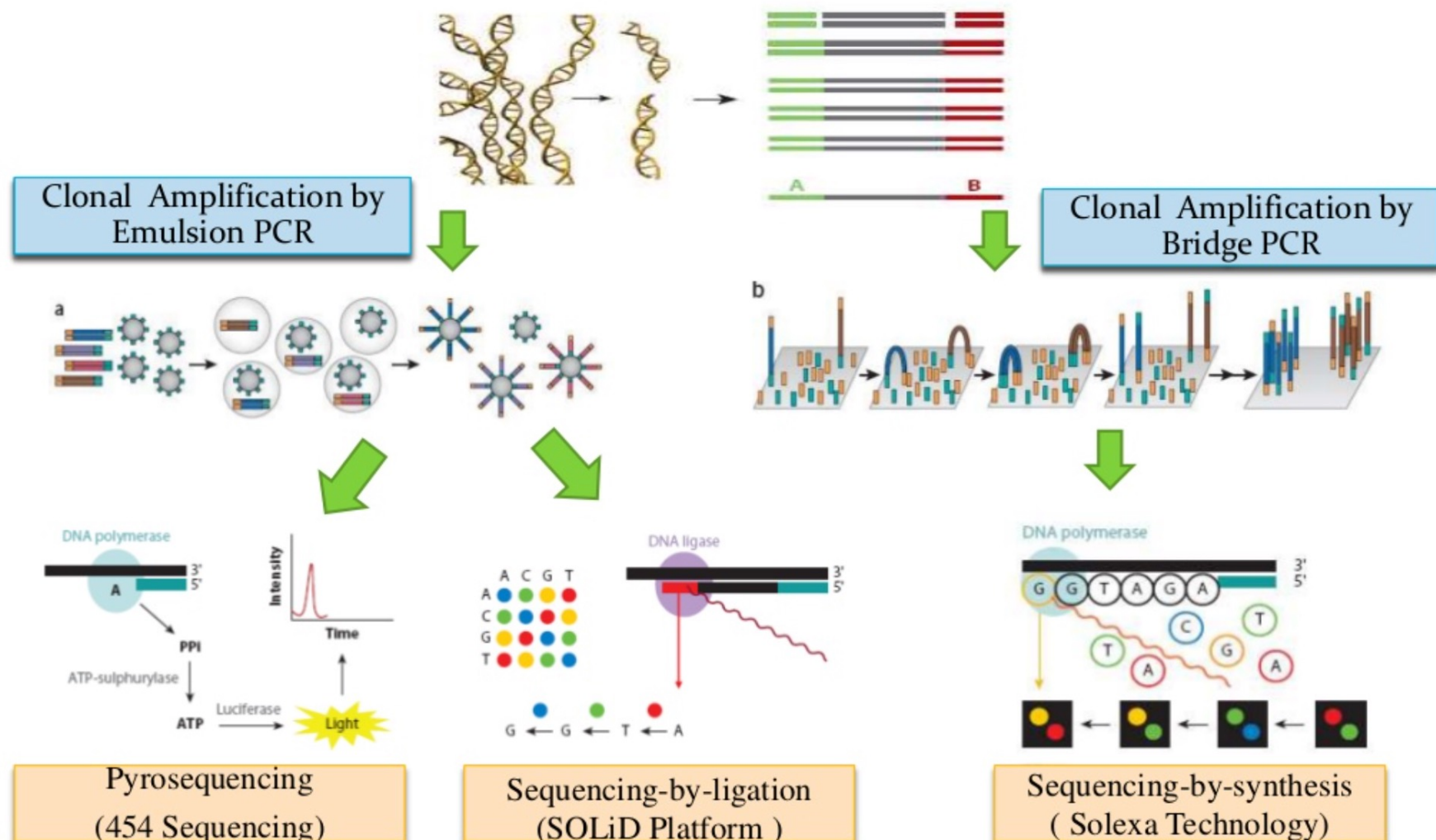
- Ion torrent sequencing



2nd-gen sequencing method (go over)

NGS WORKFLOW

Sample Extraction , DNA fragmentation and *invitro* adapter ligation



What is 3rd-gen sequencing for?

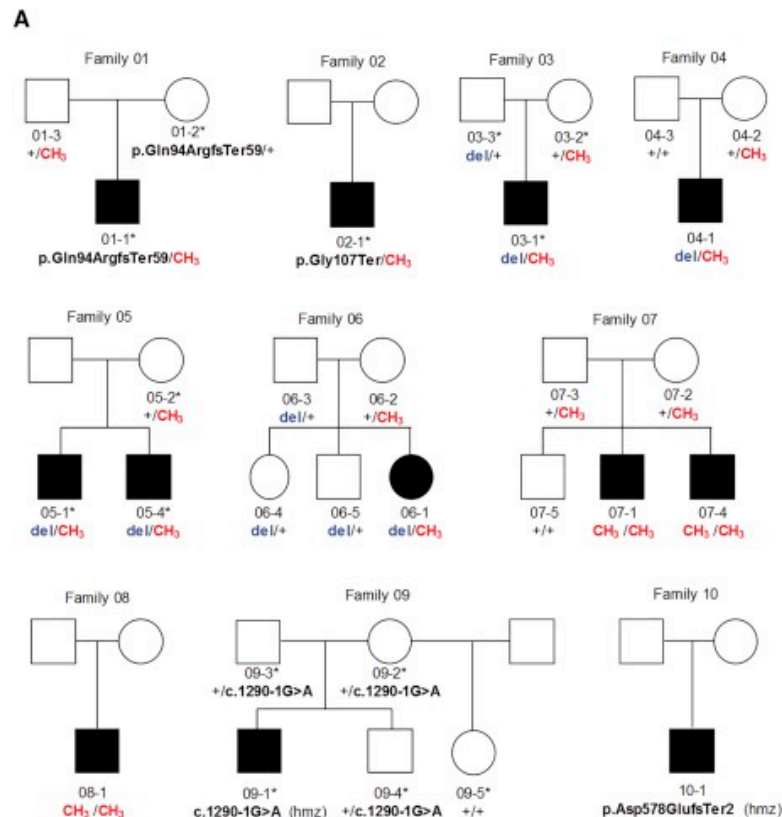
A Newly Recognized Syndrome With Characteristic Facial Features, Skeletal Dysplasia, and Developmental Delay



FIG. 1. First row: A: Patient 3 at 2 years and 10 months; B: Patient 2 at 2 years and 10 months; C: Patient 2 at 15 years and 11 months; D: Patient 1 at 24 years and 7 months. Second row: E: Patients 6 (left) and 5 (right) at 17 years and 21 years, respectively; F: Patient 7 at 13 years; and G: Patient 4 at 17 years old.

Baratela-Scott Syndrome

Proband 06-1
as an **infant**



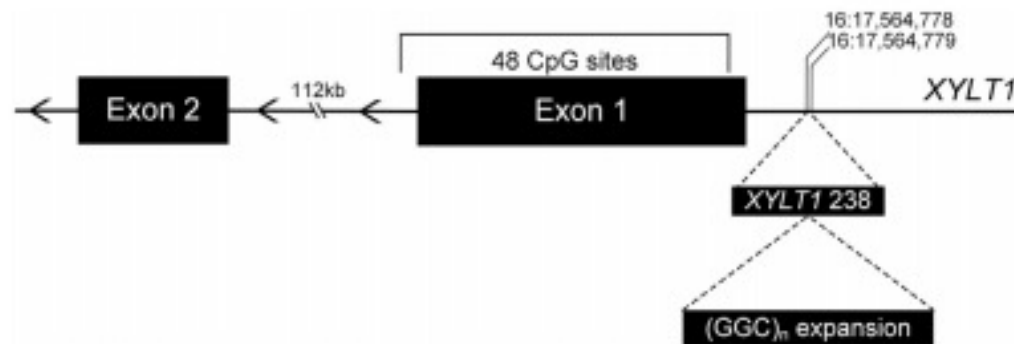
Proband 04-1 at
9 years of age

Baratela-Scott syndrome (BSS):

- a rare, autosomal-recessive disorder
- short stature, facial dysmorphisms: include flat midface, depressed nasal bridge, epicanthal folds
- developmental delay
- skeletal dysplasia
- caused by pathogenic variants in XYLT1

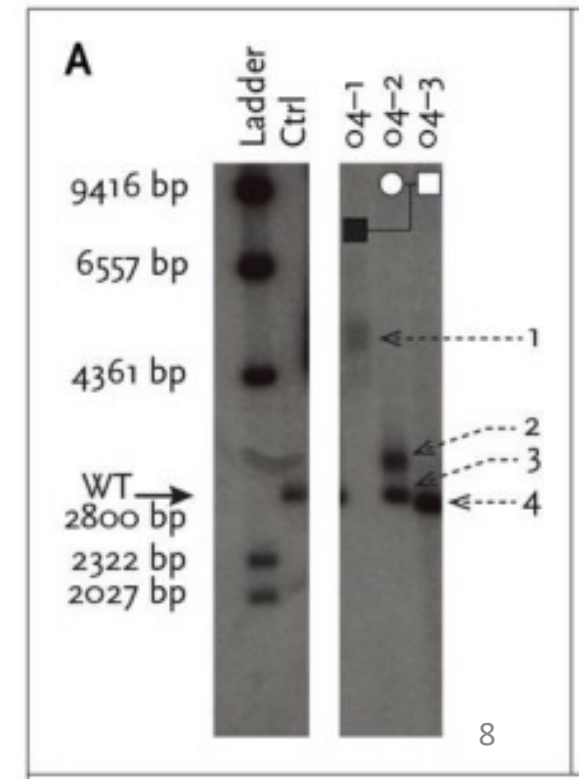
Baratela-Scott Syndrome

XYLT1 Exon 1 and Surrounding Region

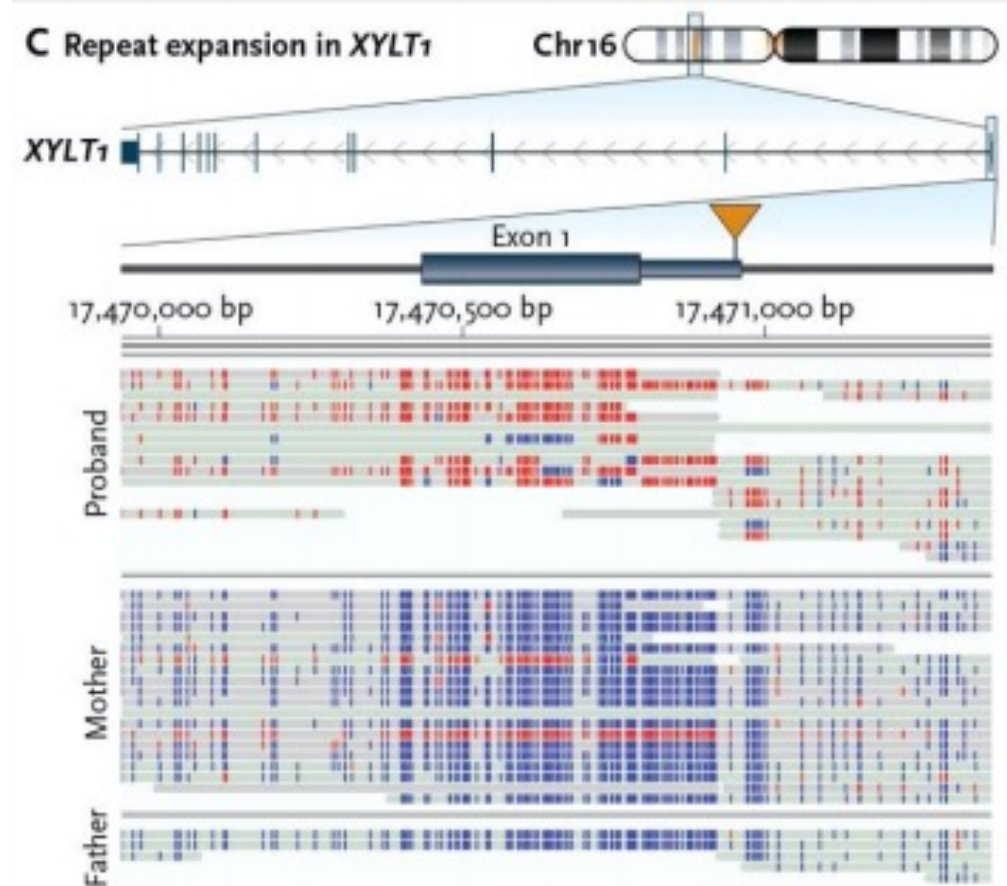
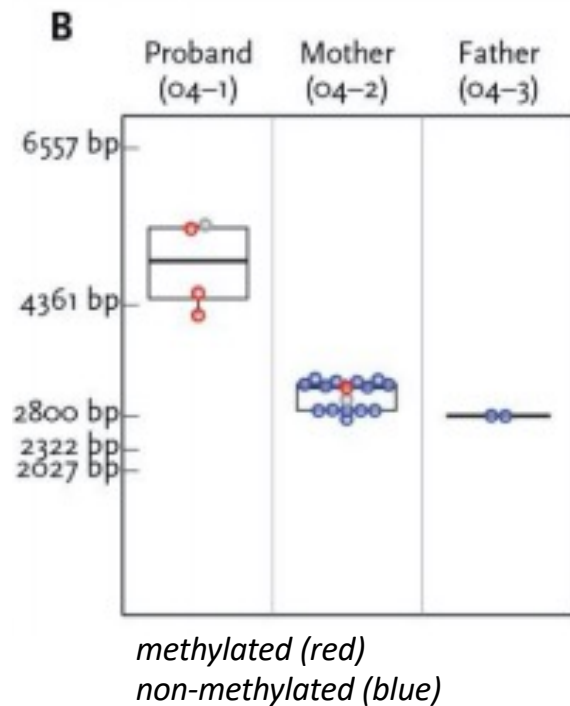


Southern blot of family 04 reported by LaCroix and colleagues :

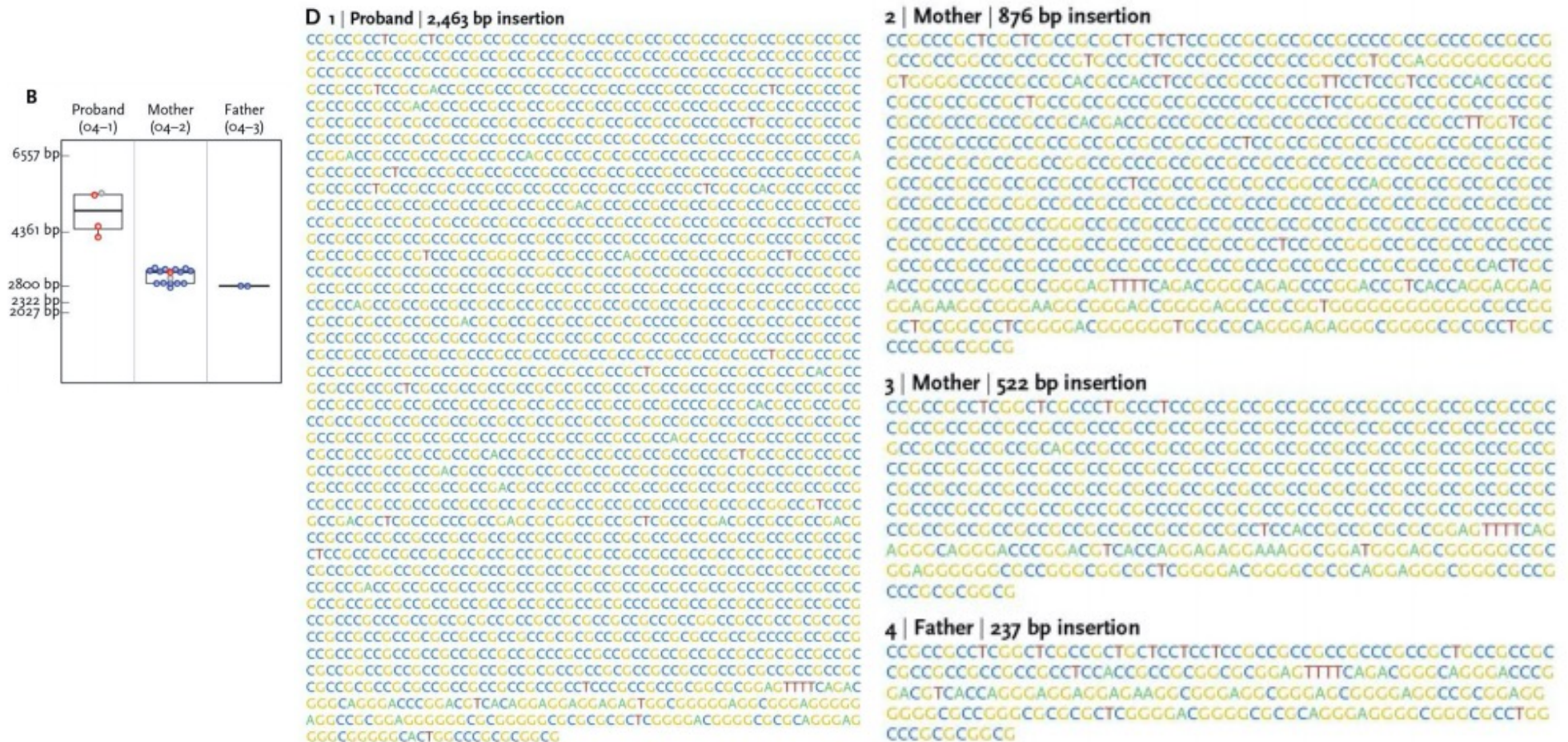
- proband (04-01) carries an expansion (1) of a region defined by two KpnI restriction enzyme sites containing a GGC repeat
- the mother (04-02) carries one premutation (2) and one wild-type allele (3)
- and the father (04-03) carries two wild-type alleles (4)



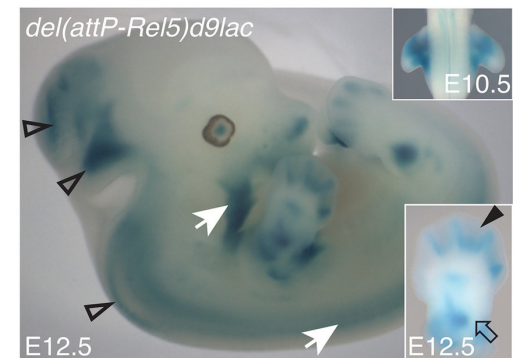
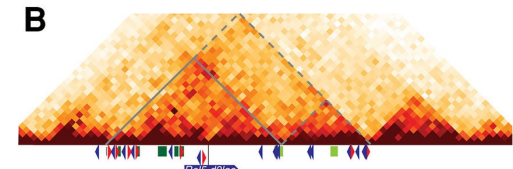
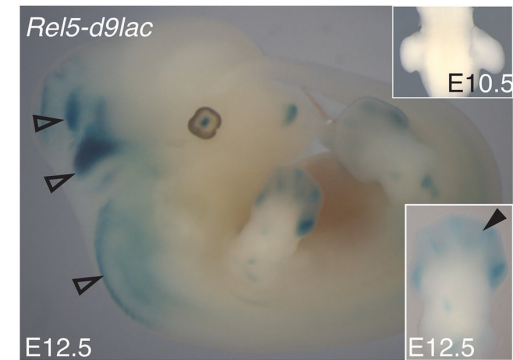
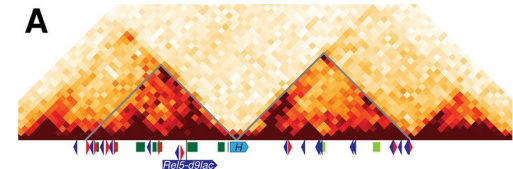
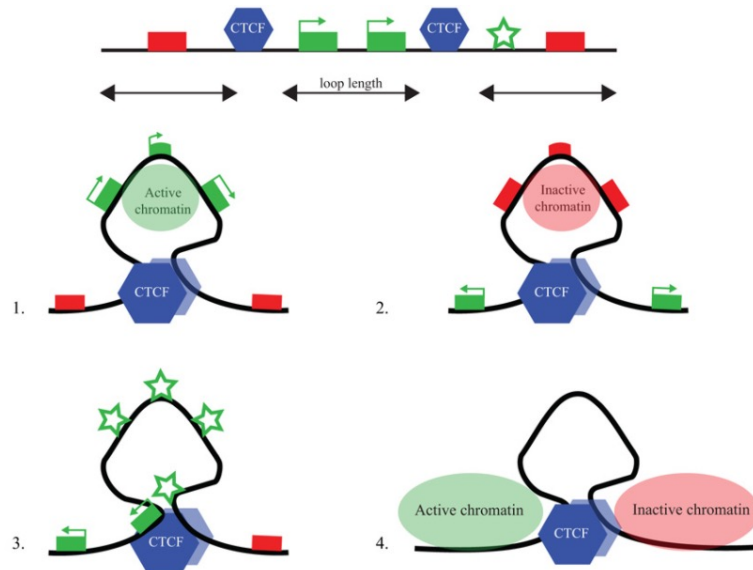
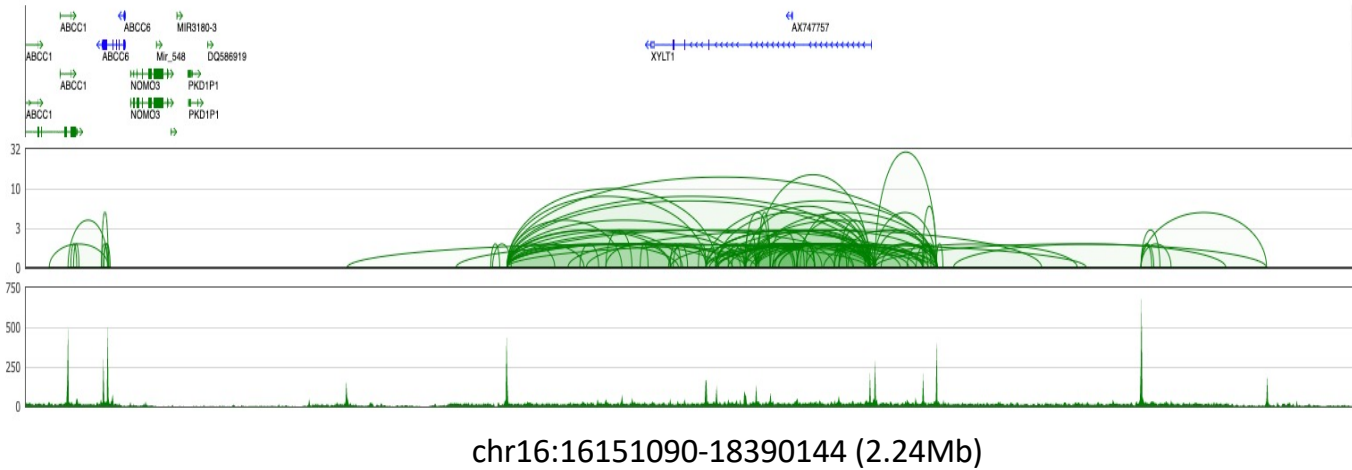
Baratela-Scott Syndrome_ Targeted long-read sequencing (T-LRS)

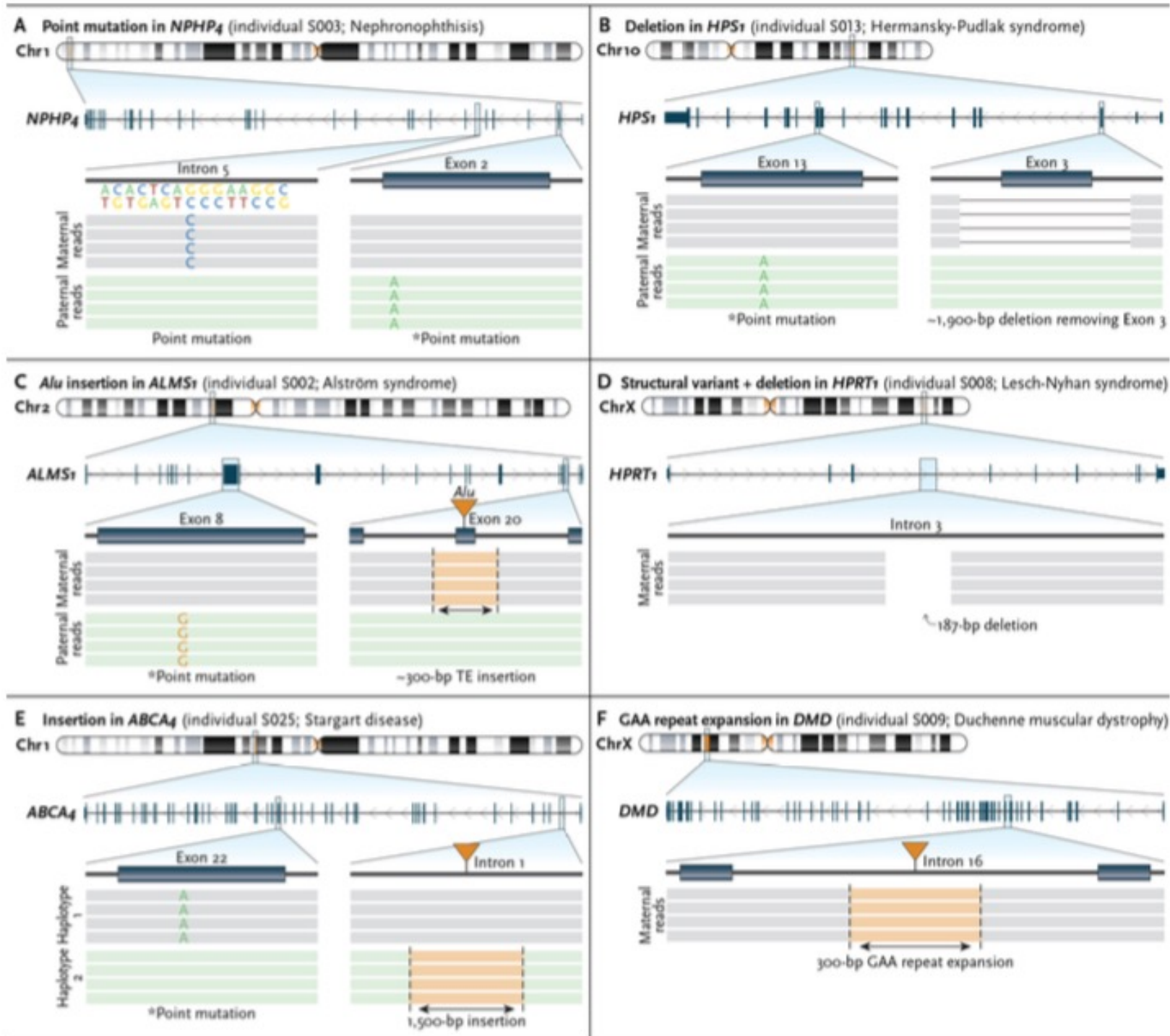


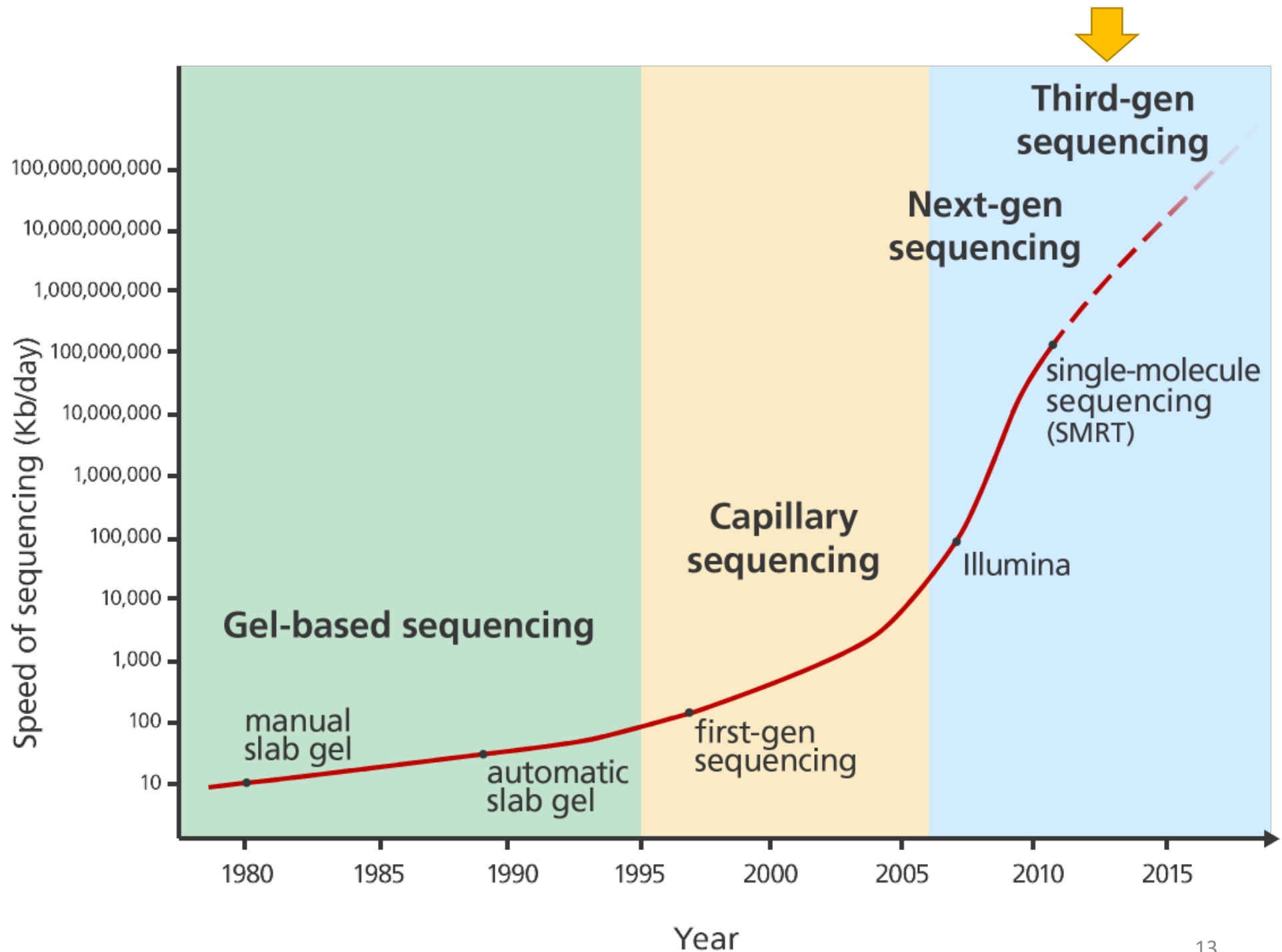
Targeted long-read sequencing (T-LRS)



XYLT1 gene region_ CTCF binding and looping information

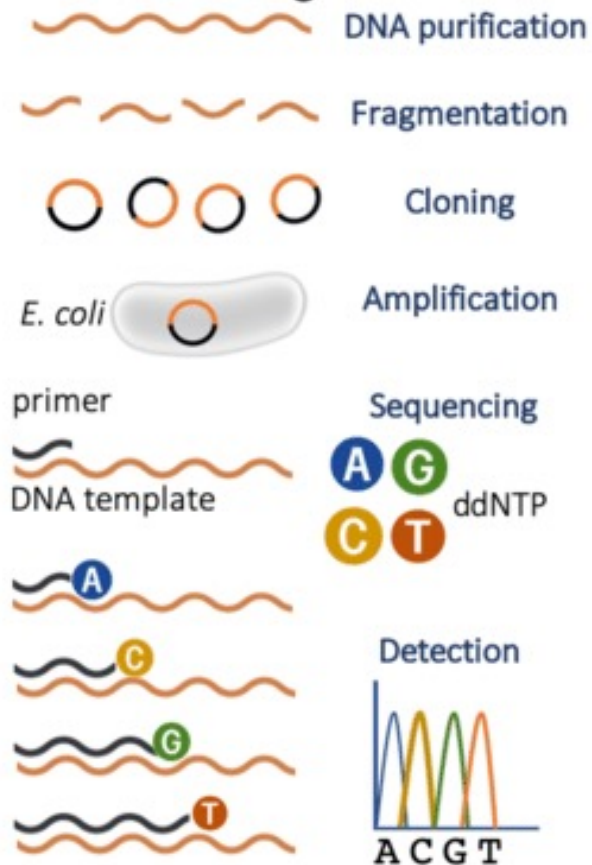






First generation sequencing (chain termination)

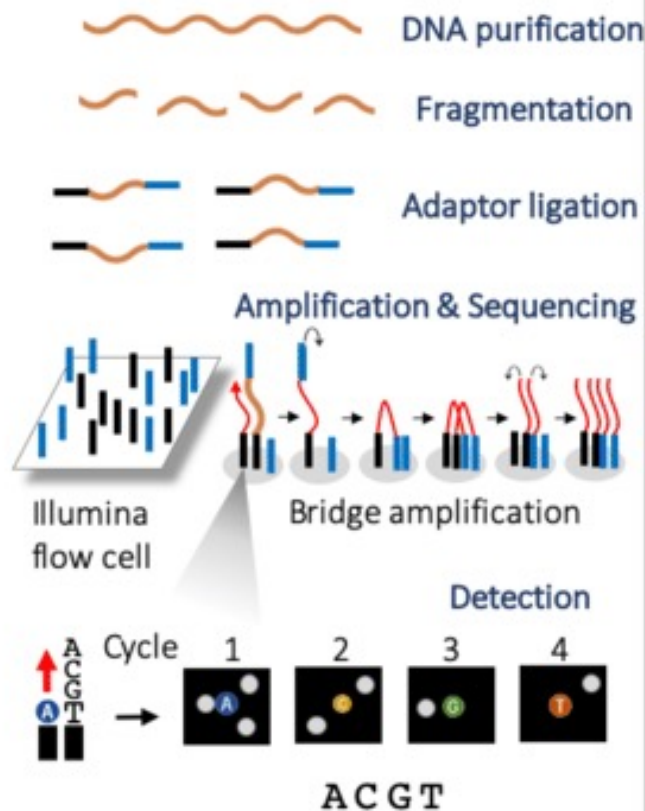
Sanger



- Sequencing by synthesis
- Read length: 300-1000 nucleotides
- Accuracy: >99%
- Real-time analysis: No
- Output data: low output

Second generation sequencing (massively parallel)

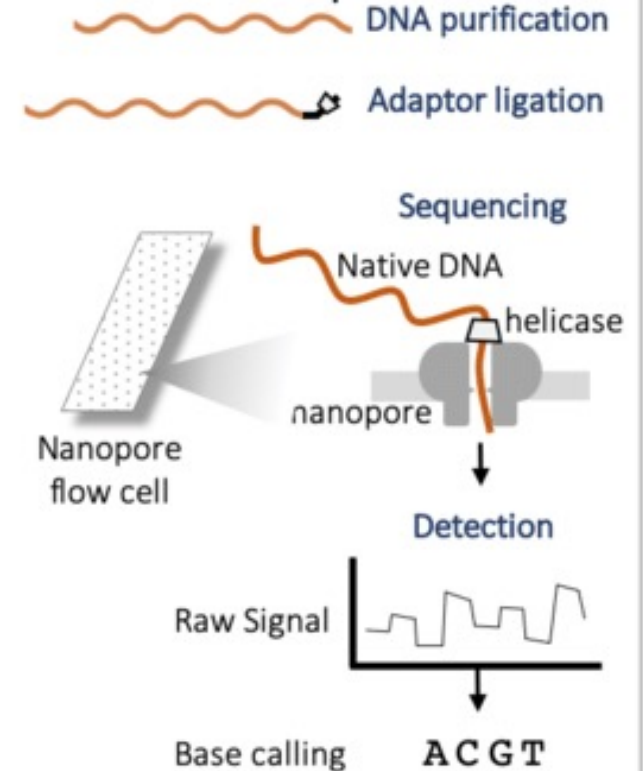
Illumina



- Sequencing by synthesis
- Read length: 36-600 nucleotides
- Accuracy: >99%
- Real-time analysis: No
- Output data: high output

Third generation sequencing (single molecule)

Nanopore



- Single molecule sequencing
- Read length: 200-2 million nucleotides
- Accuracy: ~90-95%
- Real-time analysis: Yes
- Output data: high output

Generations of Sequencers

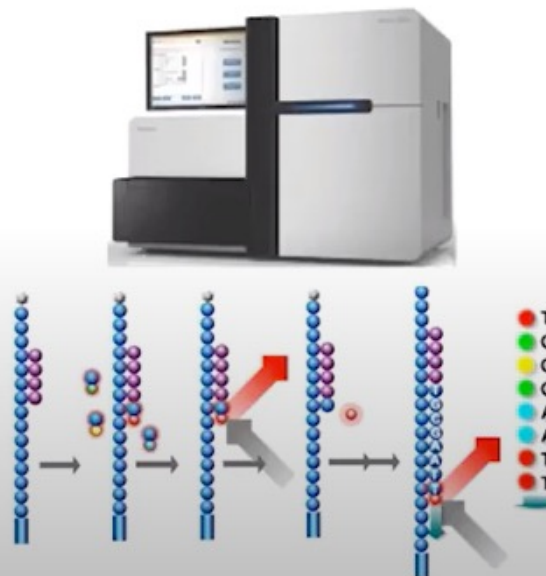
1st Generation

e.g. Sanger



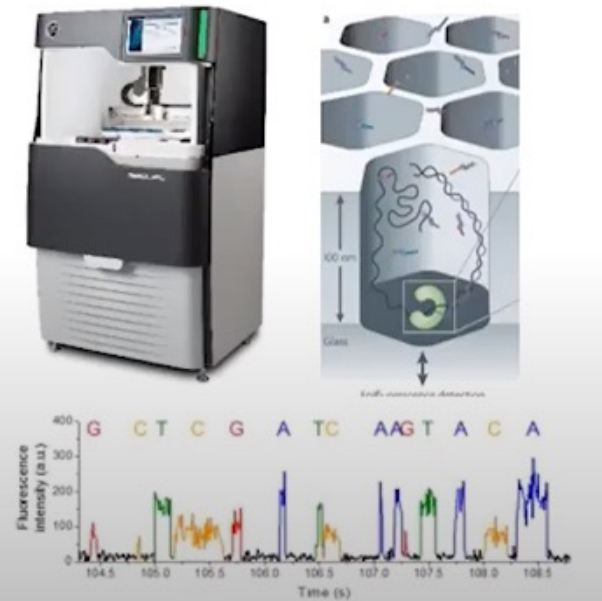
2nd Generation

e.g. Sequencing by Synthesis



3rd Generation

e.g. SMRT Sequencing



Comparison of 1st - 2nd - 3rd - gen sequencing

	First generation	Second generation	Third generation
Fundamental technology	Size-separation of specifically end-labeled DNA fragments	Wash-and-scan SBS	Single molecule real time sequencing
Resolution	Averaged across many copies of the DNA molecule	Averaged across many copies of the DNA molecule	Single DNA molecule
Current raw read accuracy	High	High	Lower
Current read length	Moderate (800-1000 bp)	Short (generally much shorter than Sanger sequencing)	> 1000 bp
Current throughput	Low	High	High
Current cost	High cost per base, Low cost per run	Low cost per base, High cost per run	Low cost per base, High cost per run
RNA-sequencing method	cDNA sequencing	cDNA sequencing	Direct RNA sequencing
Time to result	Hours	Days	< 1 day
Sample preparation	Moderately complex, PCR amplification is not required	Complex, PCR amplification is required	Various
Data analysis	Routine	Complex (due to large data volumes & short reads)	Complex
Primary results	Base calls with quality values	Base calls with quality values	Base calls with quality values

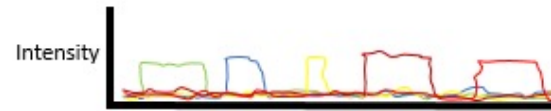
Adapted from Schadt, et al. Hum Mol Genet 2010¹³

PacBio SMRT seq

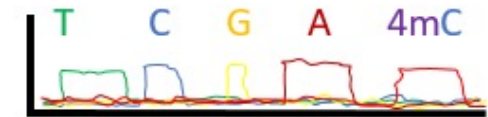
DNA passes thru polymerase in an illuminated volume



Raw output is fluorescent signal of the nucleotide incorporation, specific to each nucleotide

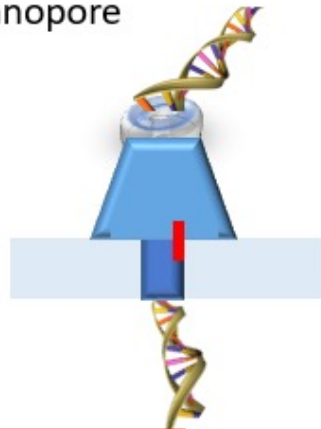


A,C,T,G have known pulse durations, which are used to infer methylated nucleotides

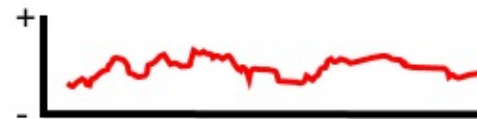


Oxford Nanopore

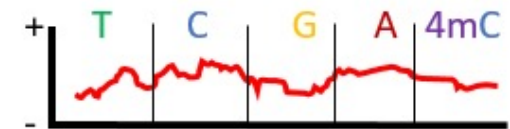
DNA passes thru nanopore



Raw output is electrical signal caused by nucleotide blocking ion flow in nanopore

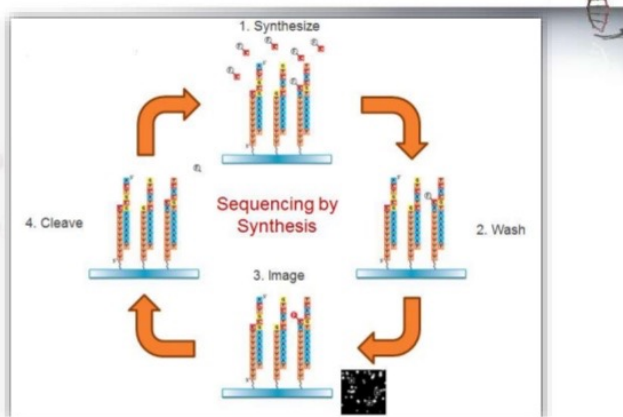


Each nucleotide has a specific electric "signature"

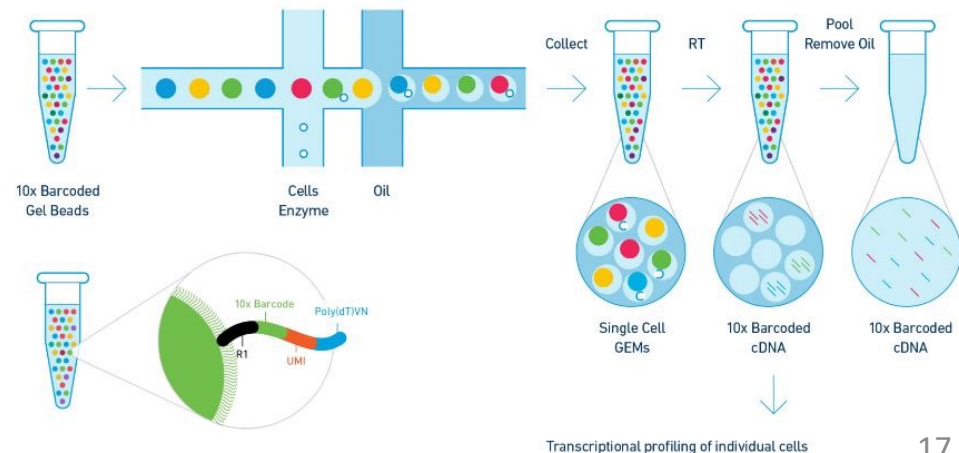


SeqLL

Sequencing by Synthesis



10x Genomics



PacBio

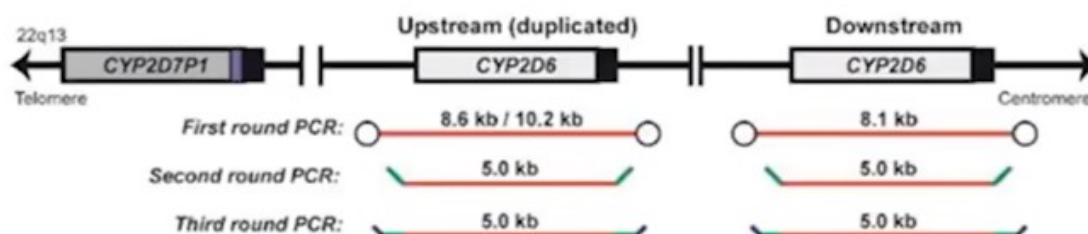
Part I

Chapter 4

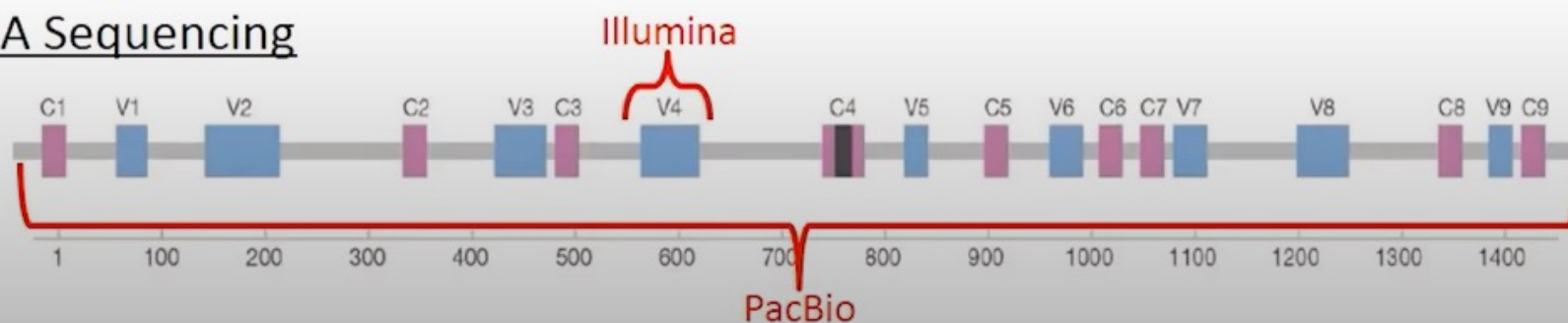
PacBio Advantages for Targeted Resequencing

PER SMRT CELL	Amplicon size	
Per Sample Diversity	<3 kb	>3kb
Low diversity (e.g., plasmids, clonal verification)	✓✓✓	✓
High diversity (16S samples, vH/vL)	✓	✗

Haplotyping/Genotyping



16S rRNA Sequencing



SHANK3 – Involved in Autism





Pacific Biosciences

- Pacific Biosciences
 - Installed summer 2013
 - Single molecule sequencing
 - Very long read lengths (up to 60 kb)
 - Rapid sequencing
 - Can detect base modifications (i.e. methylation)
 - Relatively low throughput

SMRT® Sequencing Advantages

Long
Reads

Uniform
Coverage

High
Accuracy

Single-
Molecule
Resolution

Epigenetics

Detection of base modifications

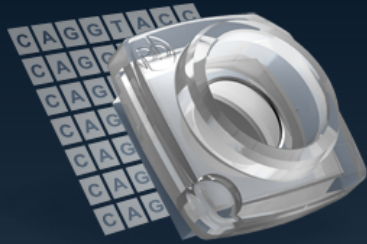


~1 bacterial genome
~1 bacterial transcriptome
1 human genome = 100 SMRT cells?

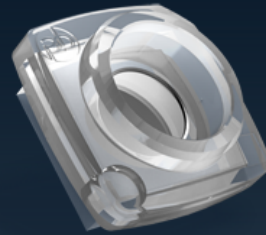


PacBio output

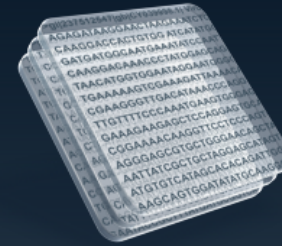
- PacBio throughput
 - ~ 1 Gb/SMRT cell
- PacBio read lengths: 500bp-60kb
- Output file format: FASTQ
- What can we use PacBio for?
 - Anything, except really large genomes



Open Existing



Create New



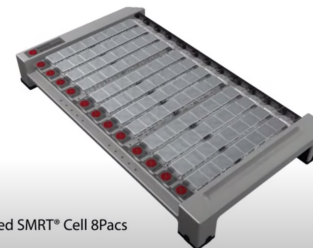
Import and Manage

RECENT JOBS

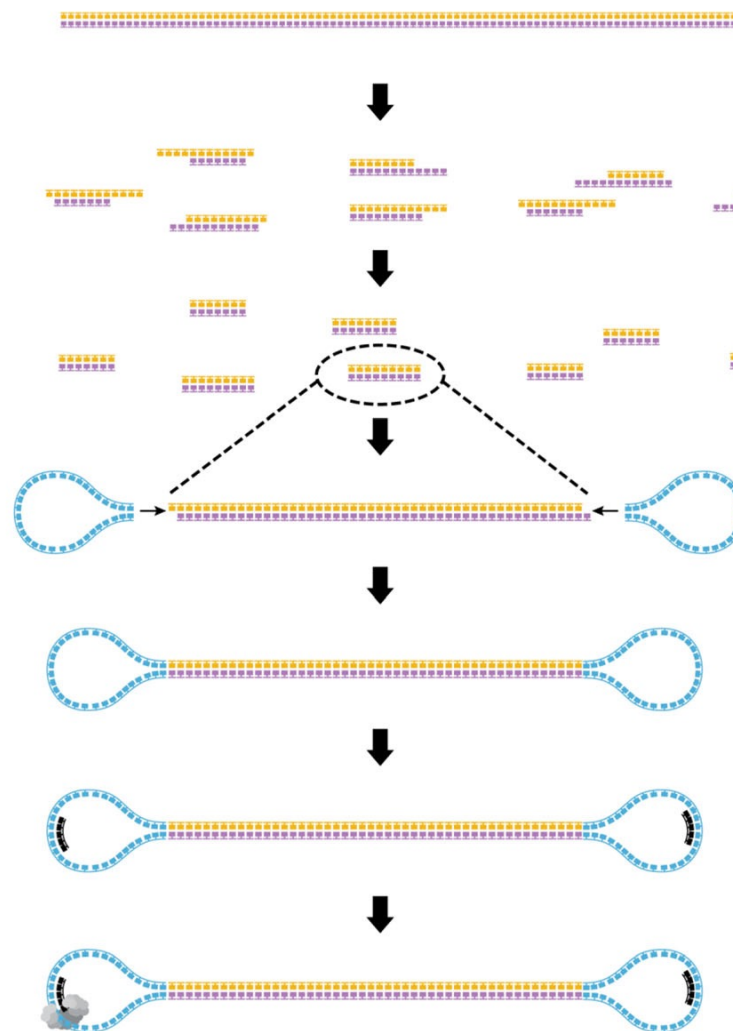
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mapping_sample_S20	RS_Resequencing.1	rcatest	2013-09-13T14:21:	Completed	ugc_admin
testaaa	RS_HGAP_Assembly.1		2013-09-13T13:29:	Completed	ugc_admin
mapping_sample_S19	RS_Resequencing.1	rcatest	2013-09-13T12:55:	Completed	ugc_admin
mapping_sample_S18	RS_Resequencing.1	rcatest	2013-09-13T11:39:	Completed	ugc_admin
sample_S3_mapping	RS_Resequencing.1	rcatest	2013-09-13T11:35:	Completed	ugc_admin



SMRT® Cell 8Pac

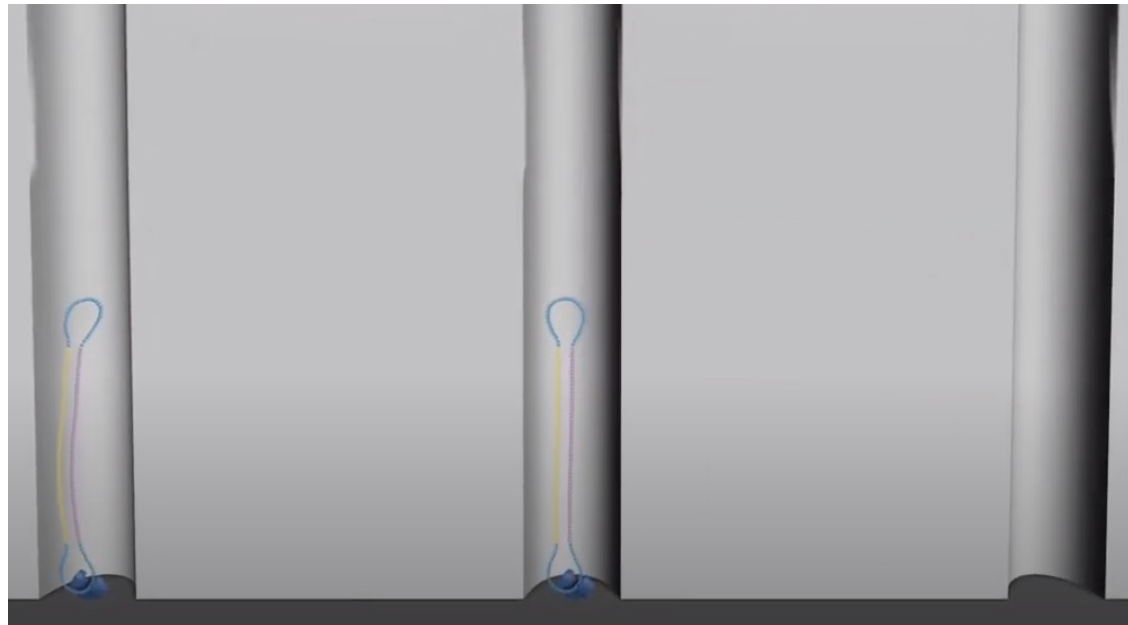
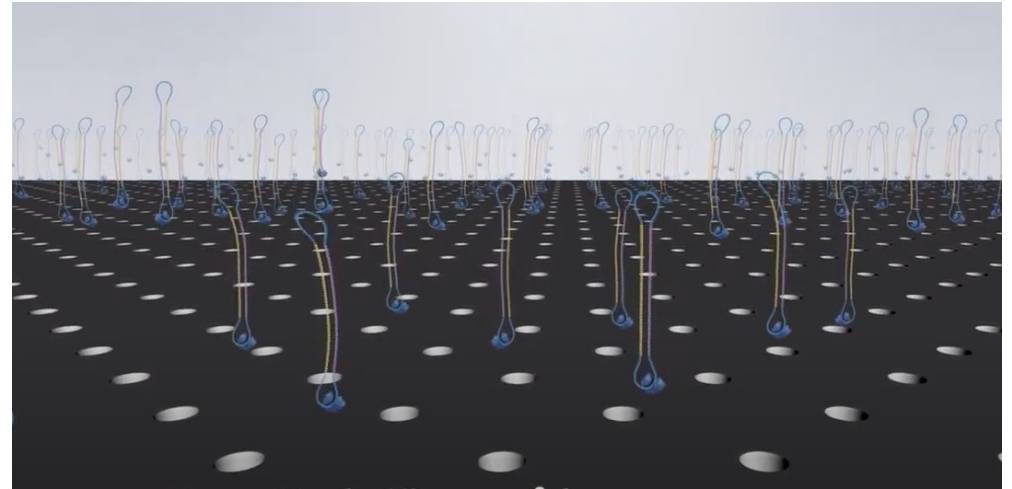
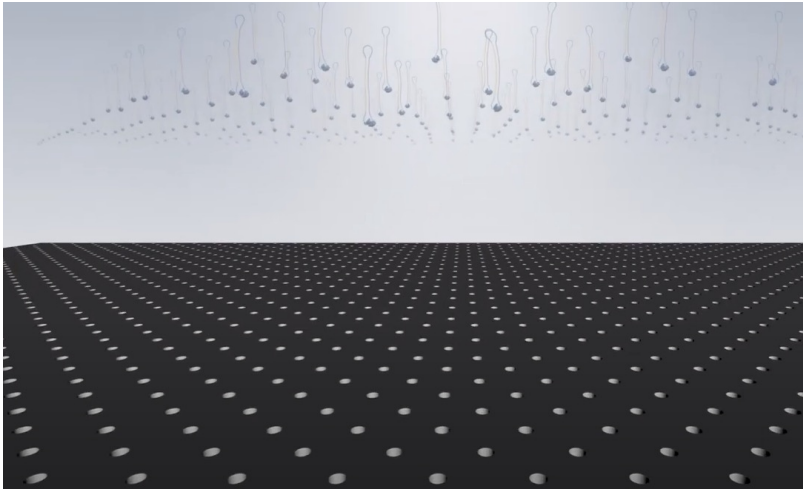


Twelve Loaded SMRT® Cell 8Pacs

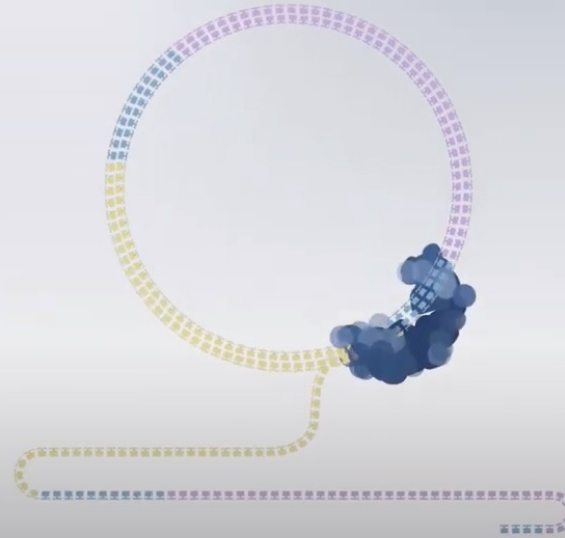


Library
construction

Library in SMRT CELL



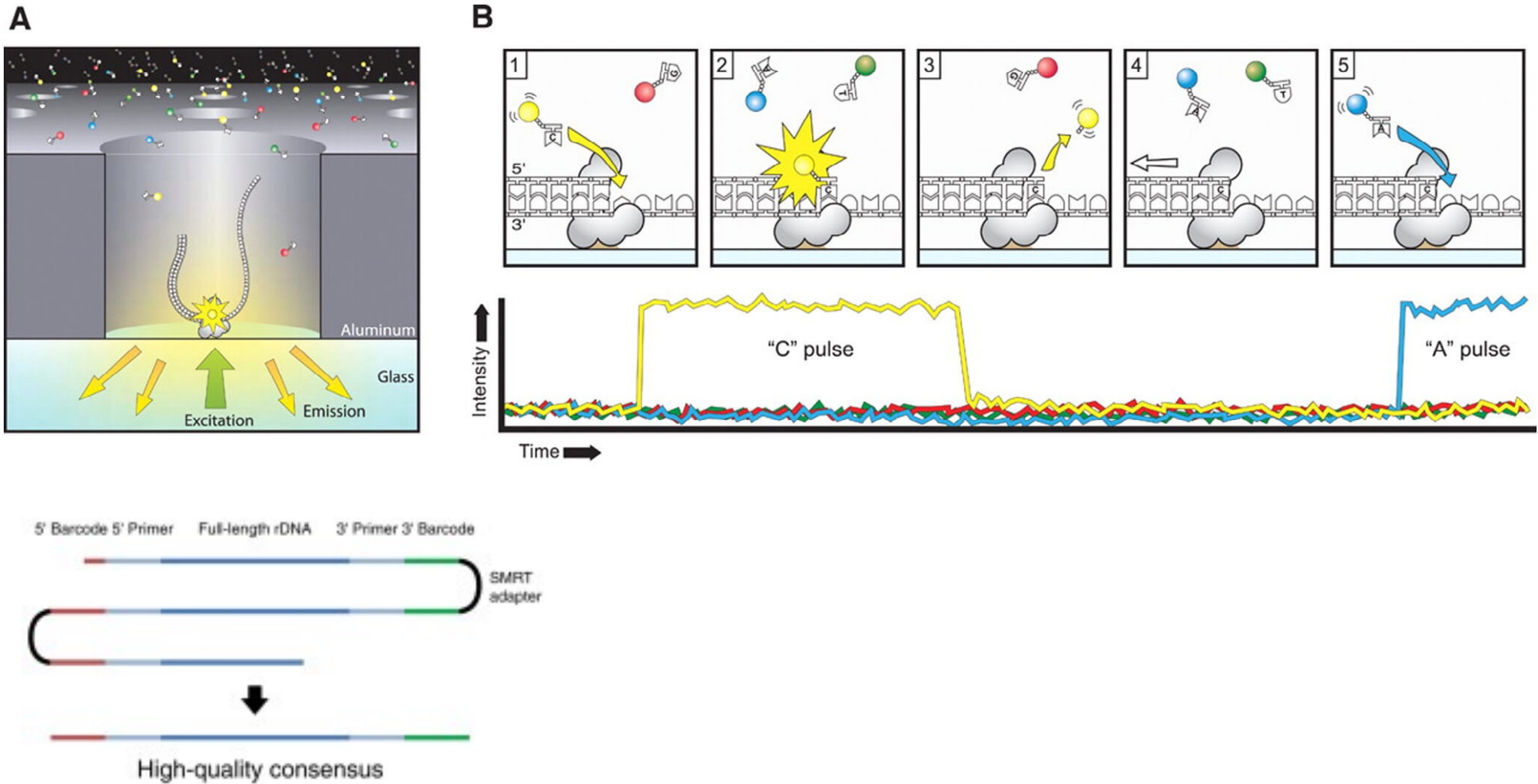
**Circular Consensus
Sequencing (CCS)
for Highly Accurate
Long Reads**



**Circular Consensus
Sequencing (CCS)
for Highly Accurate
Long Reads**



Sequencing



SMRT® Portal Home Admin Help About

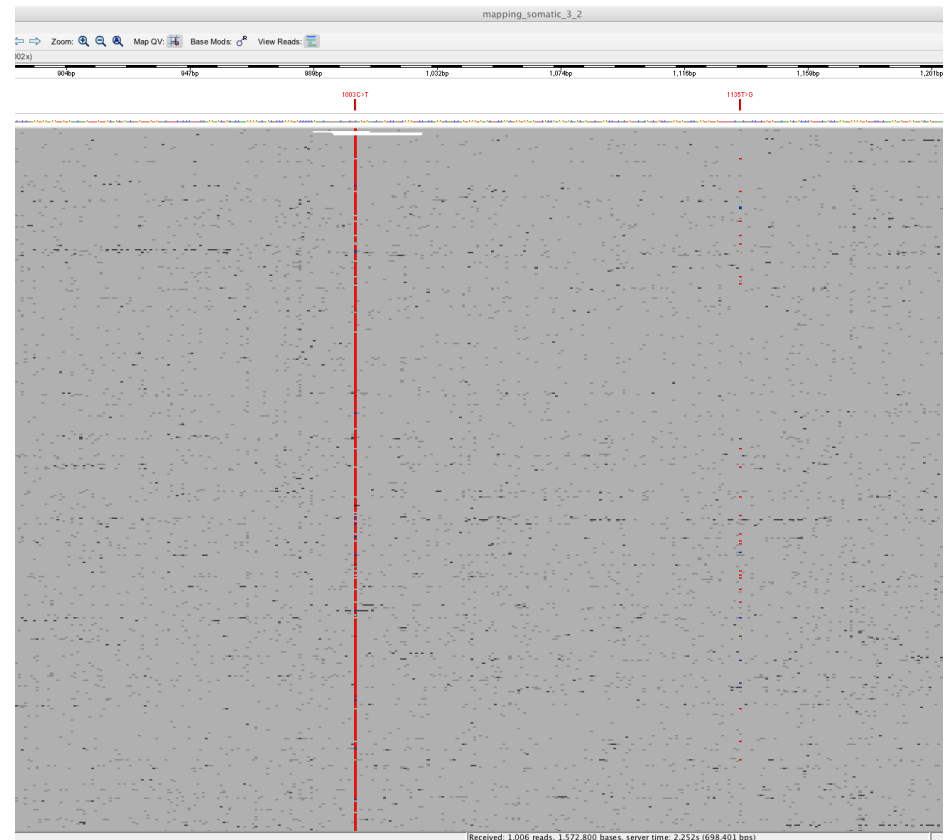
DESIGN JOB

Job Name Comments

Protocol Reference

SMRT Cells Available (Viewing 1 - 50 of 62) Search

SMRT Cell ID	Sample	V	User	Groups	Started
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0057985255000000182308820	pb_2	v2		all	2013-10
0057985255000000182308820	pb_4	v2		all	2013-10
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0057973255000000182308820	pb_1-8	v2		all	2013-09
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0057973255000000182308820	pb_1-6	v2		all	2013-09
0057973255000000182308820	pb_1-5	v2		all	2013-09



SMRT analysis pipelines

- Mapping
- Variant calling
- Assembly
- Scaffolding
- Base modifications

SMRT[®] Sequencing Applications



**WHOLE GENOME
SEQUENCING**



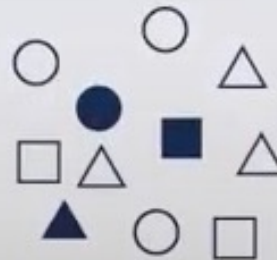
**RNA
SEQUENCING**



**TARGETED
SEQUENCING**



**VARIANT
DETECTION**



**COMPLEX
POPULATIONS**



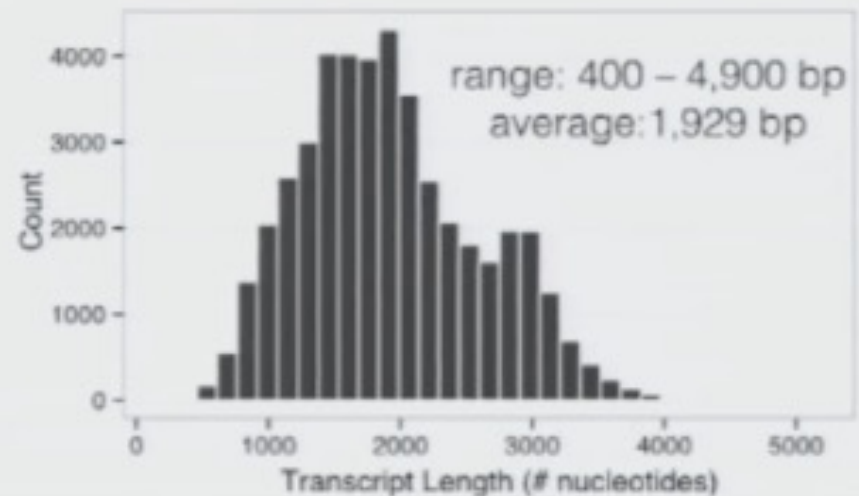
EPIGENETICS

MCF-7 Transcriptome Characterization

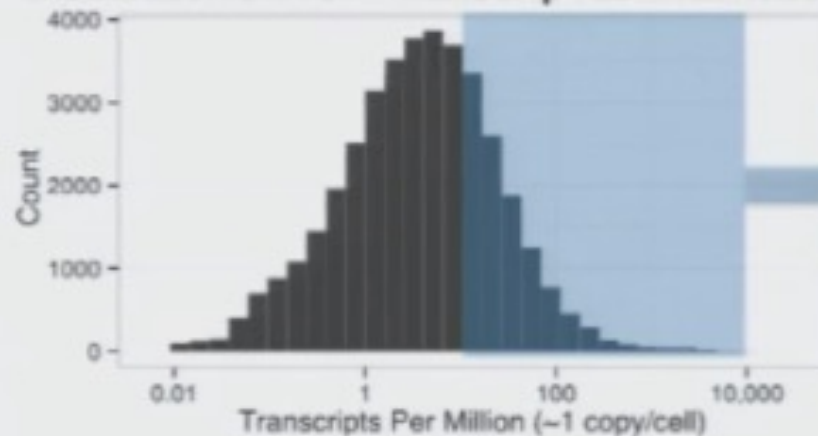
Deep dataset of full-length cDNA sequencing of RNA from human breast cancer cell line MCF-7.
~14 million full-length mRNA reads

- 44,531 transcript isoforms
- 14,385 genes
- +93 gene fusions

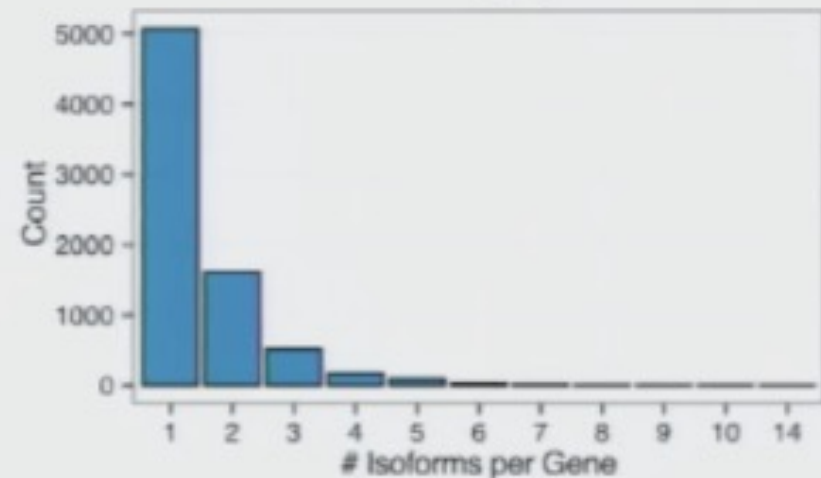
Distribution of detected transcript lengths



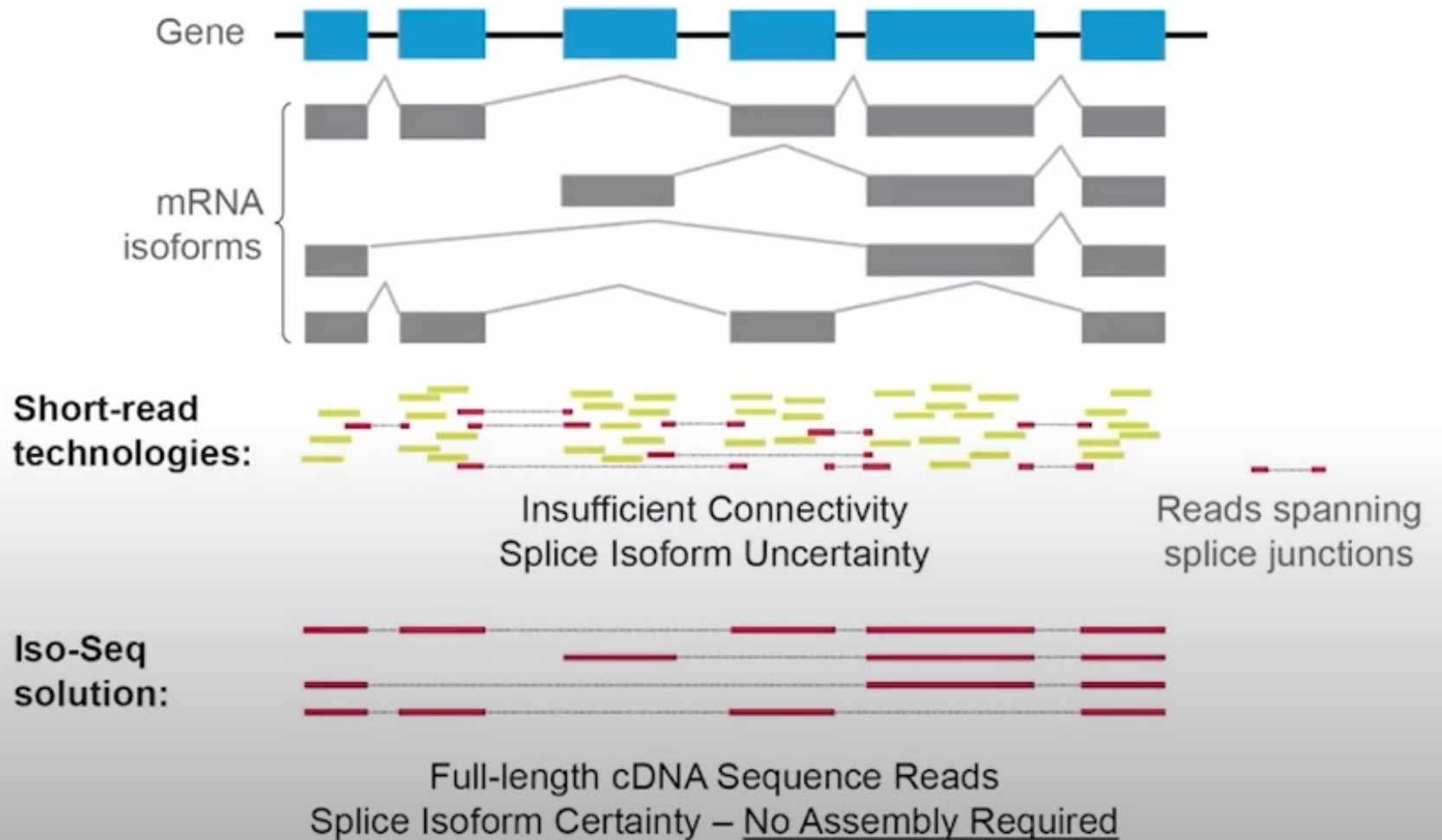
Distribution of MCF7 transcript abundances



Number of genes producing multiple distinct transcript products



PacBio for Full length mRNA Sequencing—Iso-Seq

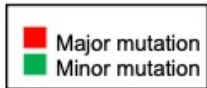


A

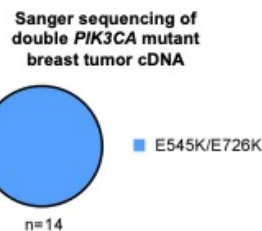
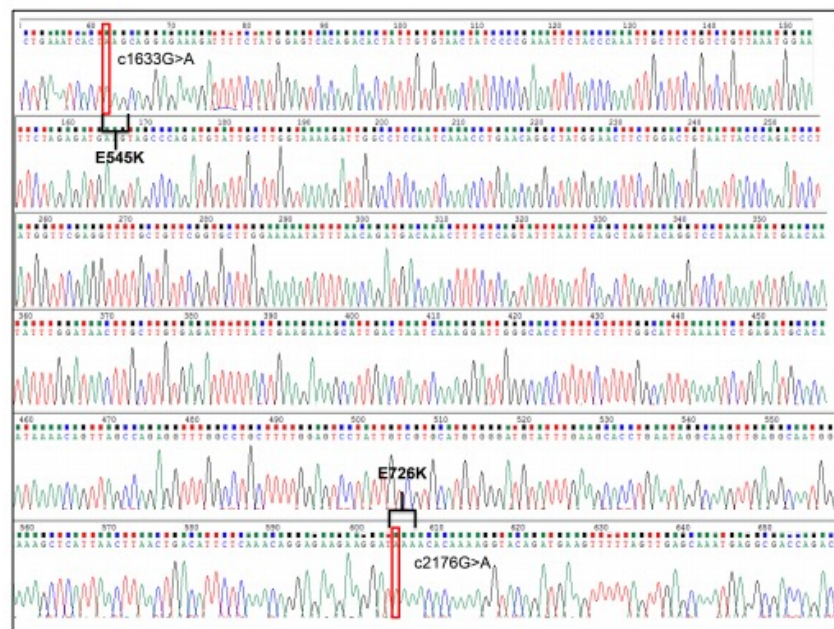
FFPE archival tumor sample

Fresh tumor sample

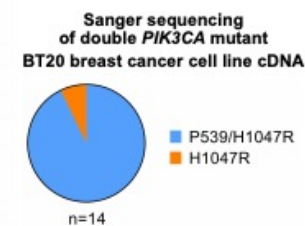
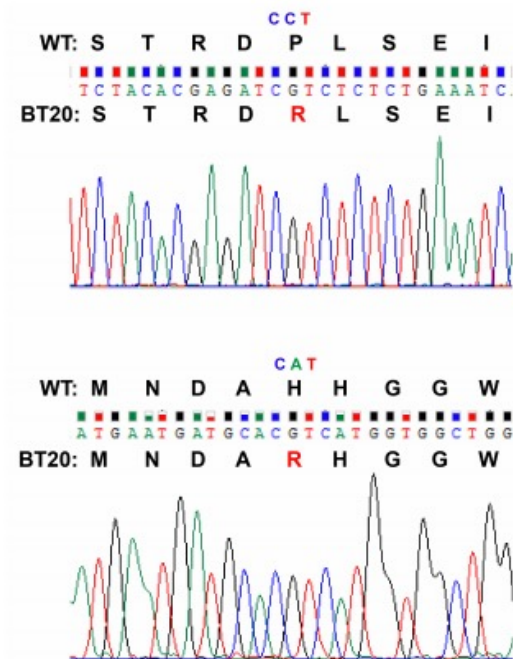
Double <i>PIK3CA</i> mutants	gDNA distance	Resolvable by NGS	cDNA distance	Resolvable by NGS	Resolvable by Sanger-seq	Resolvable by SMRT-seq
E542K + E453Q/K	8.00 kb	No	0.27 kb	Yes	Yes	Yes
E542K + E726K	2.85 kb	No	0.56 kb	No	Yes	Yes
E542K + M1043L/I	16.00 kb	No	1.51 kb	No	No	Yes
E545K + E453Q/K	8.01 kb	No	0.28 kb	Yes	Yes	Yes
E545K + E726K	2.84 kb	No	0.55 kb	No	Yes	Yes
E545K + M1043L/I	15.98 kb	No	1.50 kb	No	No	Yes
H1047R + E453Q/K	24.01 kb	No	1.79 kb	No	No	Yes
H1047R + E726K	13.15 kb	No	0.97 kb	No	No	Yes



B



C



RESEARCH ARTICLE

Double *PIK3CA* mutations in cis increase oncogenicity and sensitivity to PI3Kα inhibitors

Neil Vasan^{1,2,3}, Pedram Razavi^{1,2,*}, Jared L. Johnson^{3,*}, Hong Shao^{1,*}, Hardik Shah⁴, Alesia Antoine⁴, Erik...

+ See all authors and affiliations
 Science 08 Nov 2019;
 Vol. 366, Issue 6466, pp. 714-723
 DOI: 10.1126/science.aaw9032

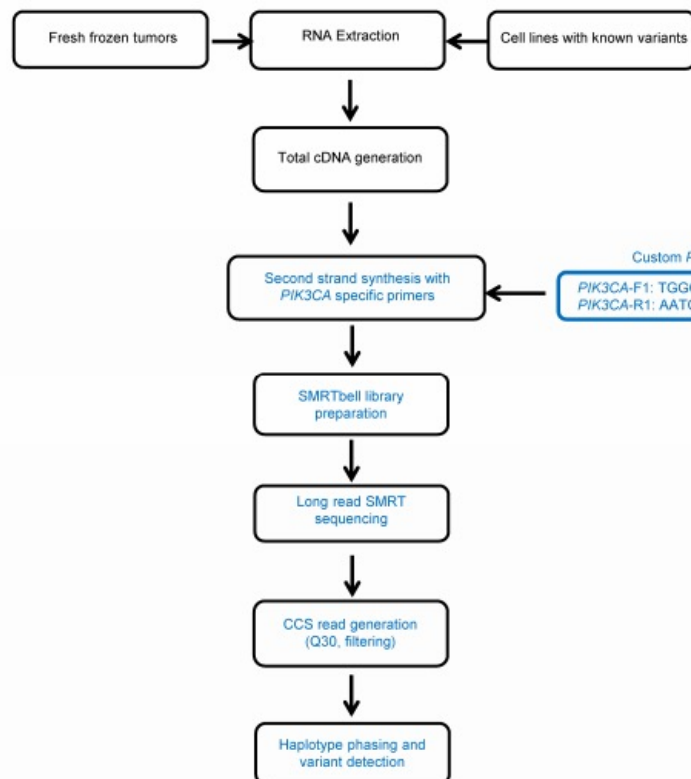


Science

Vol 366, Issue 6466
08 November 2019

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[Classified \(PDF\)](#)
[Masthead \(PDF\)](#)

D

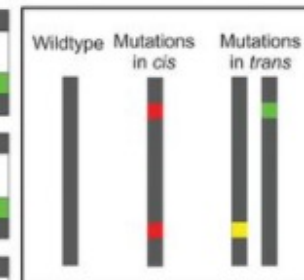


Custom PIK3CA primers

PIK3CA-F1: TGGGACCCGATGCGGTTA
PIK3CA-R1: AATCGGTCTTTGCCTGCTGA

SMRT-seq of breast tumors

Sample	PIK3CA			Sample Variants			Haplotype %			
	Codon	AA	Position	AA	Codon	%	A	B	C	D
E545K/E726K	GAG	E	545	K	AAG	57	58.1	39.5	2.2	0.2
	GAA	E	726	K	AAA	59				
						14731				
E545K/T1025A	GAG	E	545	K	AAG	46	54.4	45.1	0.3	0.1
	ACC	T	1025	A	GCC	45				
						20520				
E545K/M1043L	GAG	E	545	K	AAG	62	62.8	35.7	0.8	0.7
	ATG	M	1043	L	TTG	63				
						21161				
E453K/H1047R	GAA	E	453	K	AAA	95	95.8	4.1	0.2	0.1
	CAT	H	1047	R	CGT	95				
						30989				
E542K/E726K	GAA	E	542	K	AAA	43	57.7	41.8	0.3	0.1
	GAA	E	726	K	AAA	41				
						20941				
P539R/H1047R	CCT	P	539	R	CGT	29	62.8	23.3	7.6	6.3
	CAT	H	1047	R	CGT	31				
						21239				



RESEARCH ARTICLE

Double *PIK3CA* mutations in cis increase oncogenicity and sensitivity to PI3Kα inhibitors

Neil Vasan^{1,2,3}, Pedram Razavi^{1,2,*}, Jared L. Johnson^{3,*}, Hong Shao^{1,*}, Hardik Shah⁴, Alesia Antoine⁴, Erik...

+ See all authors and affiliations

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DOI: 10.1126/science.aaw9032



Science
Vol 366, Issue 6466
08 November 2019

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[Masthead \(PDF\)](#)

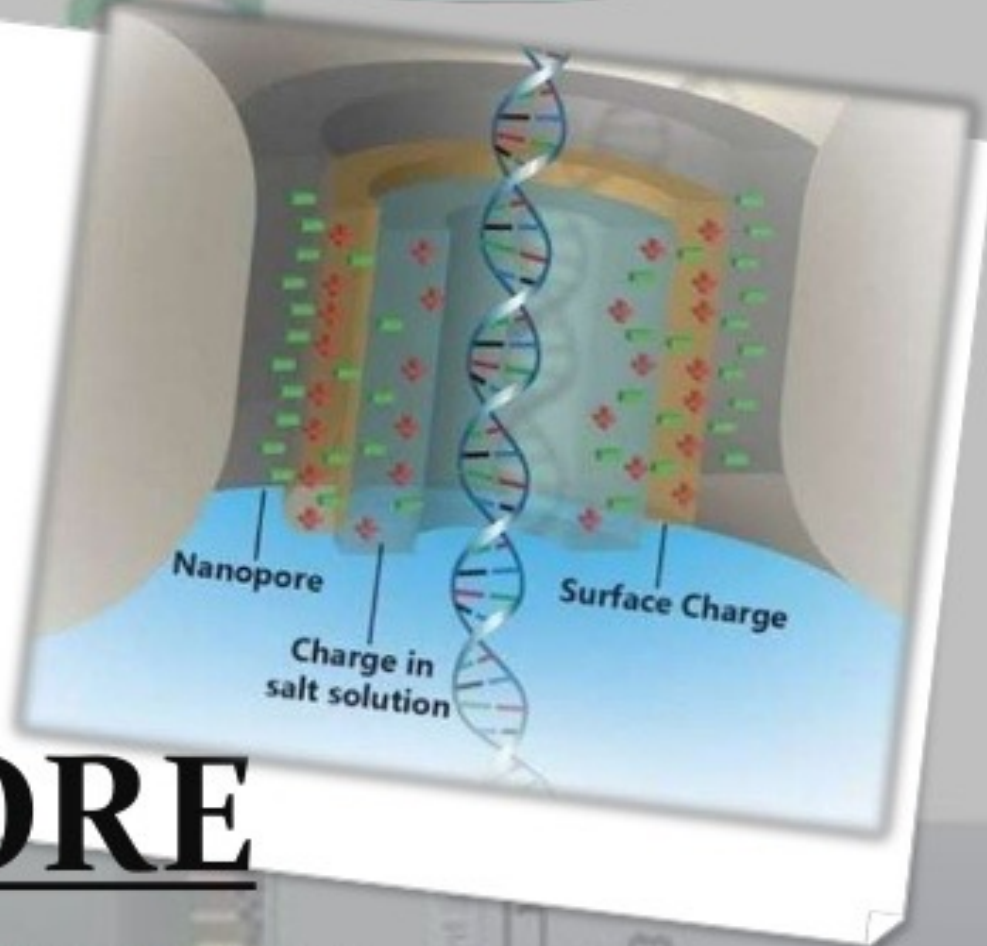
Nanopore sequencing

Part II

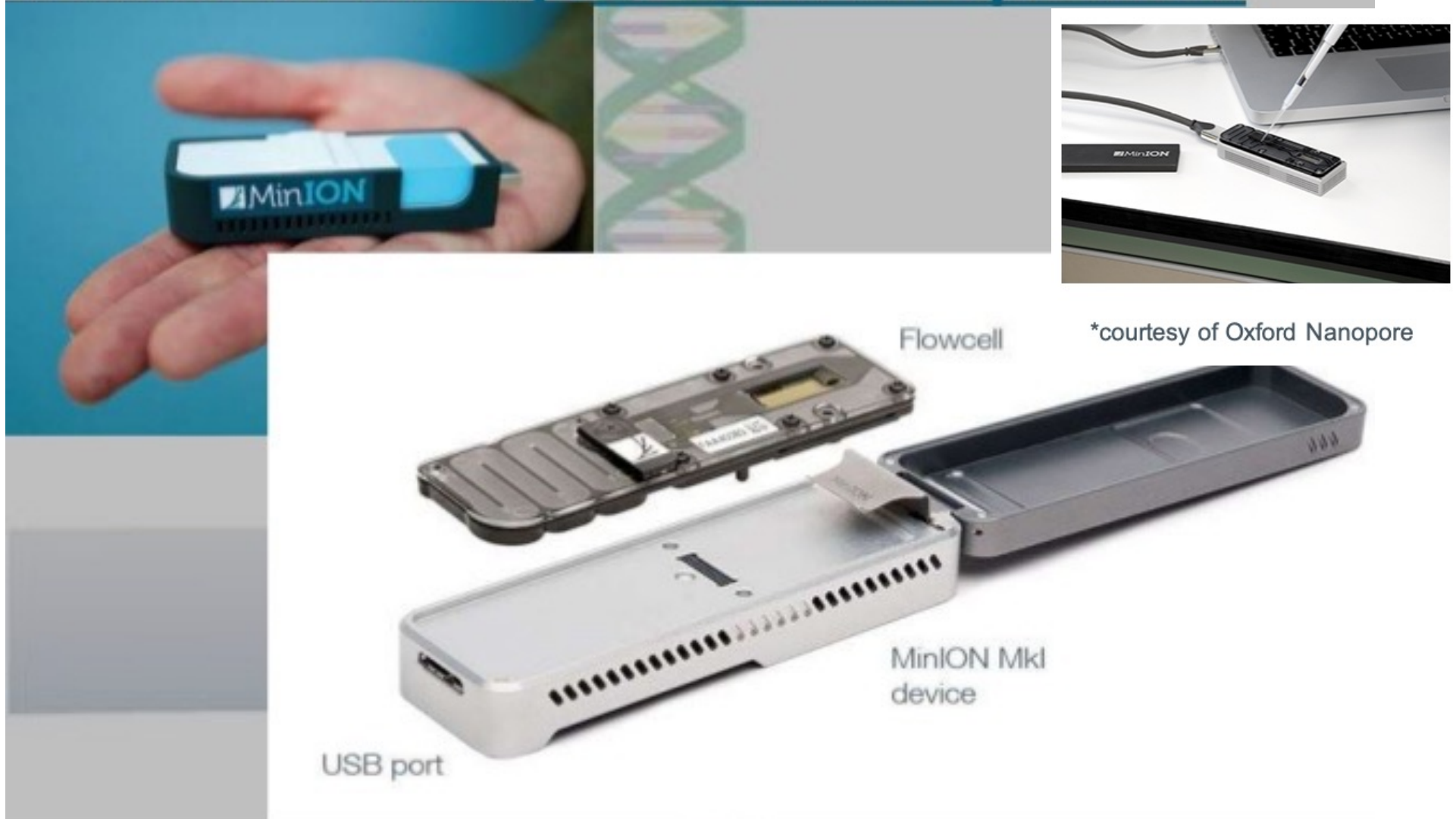
Chapter 4



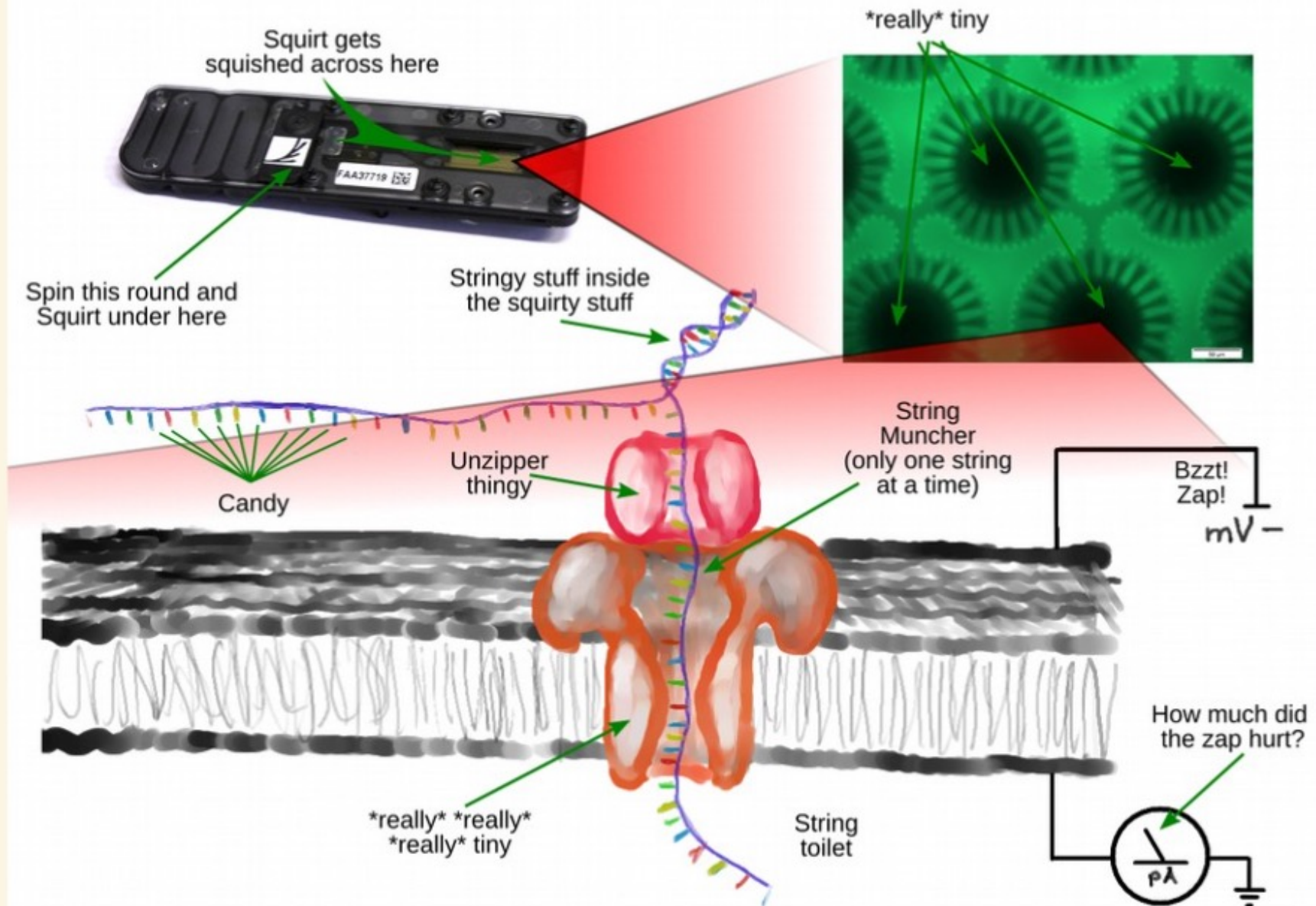
NANOPORE SEQUENCING



The portable MinION — the first handheld nanopore DNA sequencer



Nanopore Overview



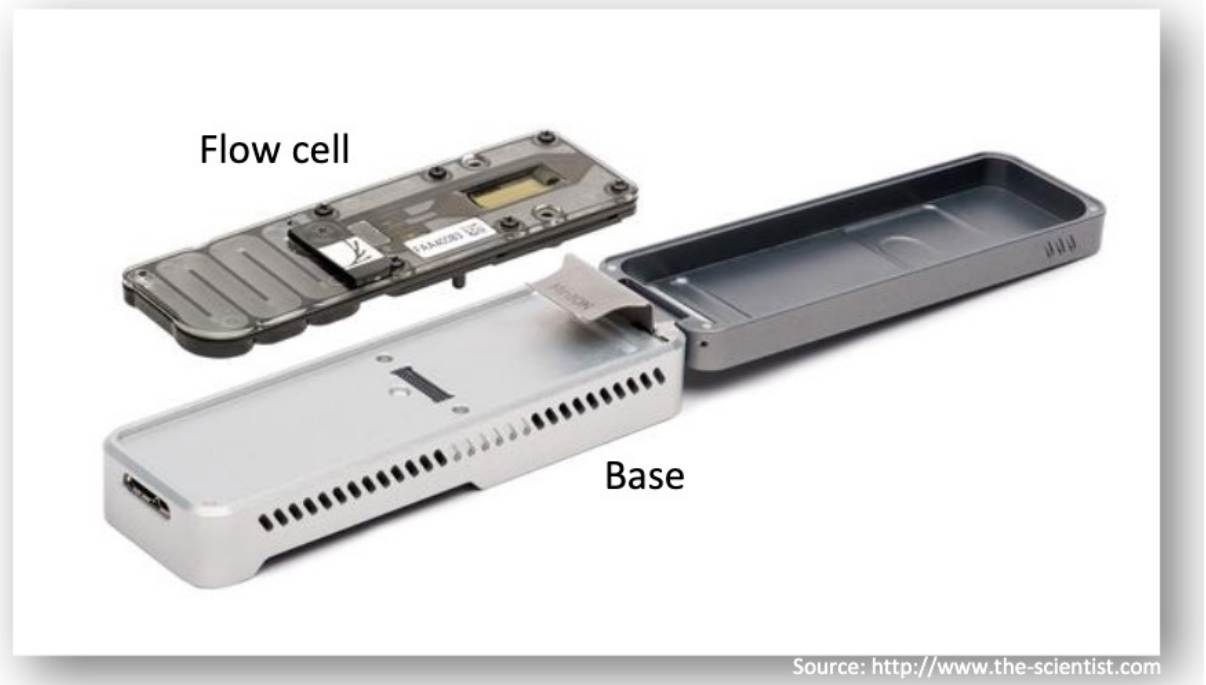
MinION Sequencer

Base

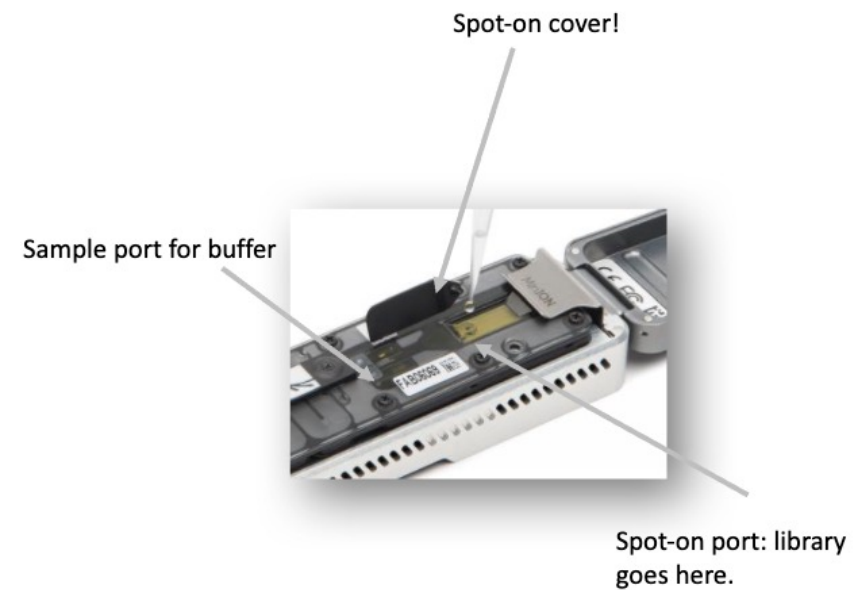
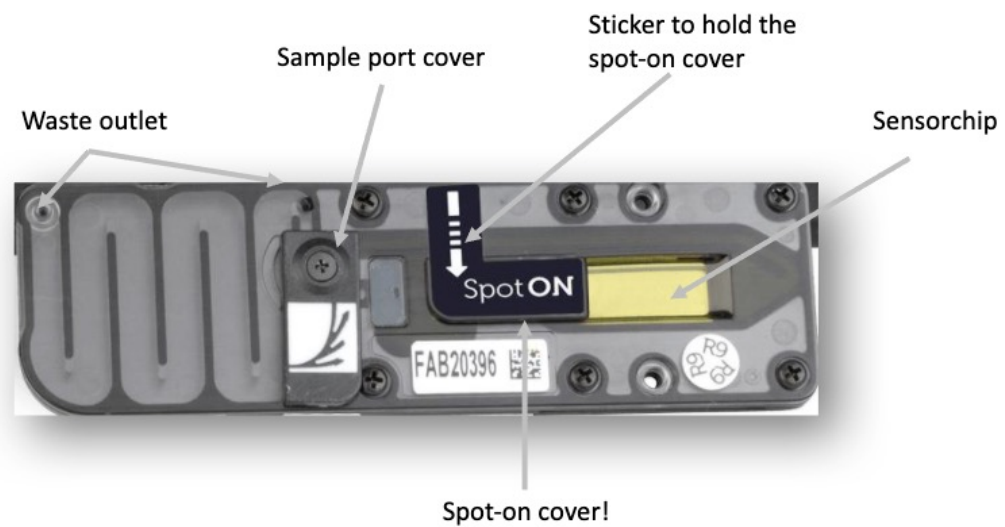
- Connects to Computer
- Fan / thermal regulation

Flow Cell

- Sample goes here
- Includes chip with nanopores
- Enables up to 48h of sequencing (Consumable)



Flow cell – Overview



A broad range of devices is now on the market – but one core technology



MinION
Mk1C

Commercially available

Complete sequencing,
analysis, and viewing device

Up to 30 Gb data / flow cell
512 channels*



MinION
Mk1B

Commercially available

Portable, USB powered
biological analysis

Up to 30 Gb data / flow cell
512 channels*



GridION
Mk1

Commercially available

Five flow cell capacity and
integrated computing

Up to 150 Gb (5 x 30 Gb) data / device
5 x 512 channels*



PromethION
P24 P48

Commercially available

High-throughput, versatile benchtop
system (P24 or P48)

P24

>3.5 Tb data /
device

24 x 3,000
channels*

P48

>7 Tb data /
device

48 x 3,000
channels*

Flongle

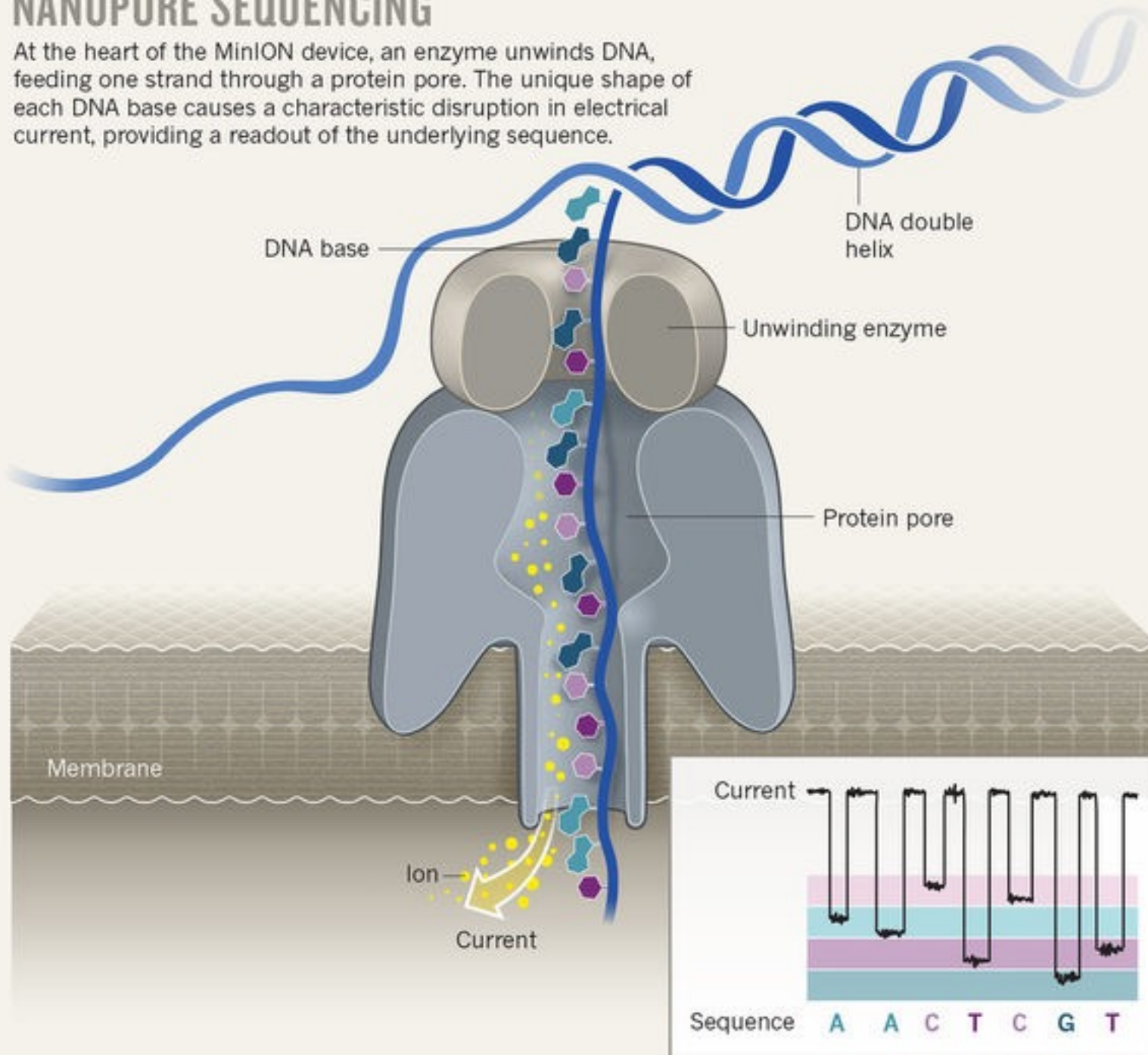
Adapter for MinION/GridION,
supports smaller single-use flow cells.
Up to 1.8 Gb currently; towards 3 Gb.



*up to this number of nanopore channels may be available for sequencing at any time

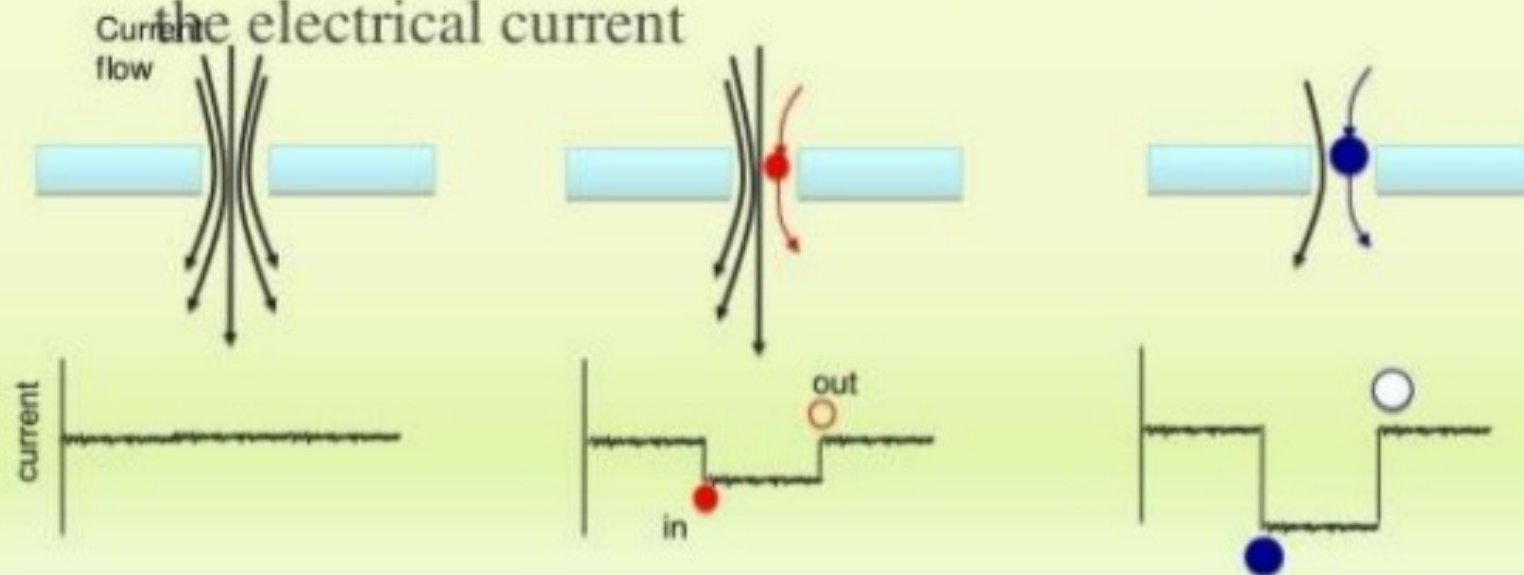
NANOPORE SEQUENCING

At the heart of the MinION device, an enzyme unwinds DNA, feeding one strand through a protein pore. The unique shape of each DNA base causes a characteristic disruption in electrical current, providing a readout of the underlying sequence.



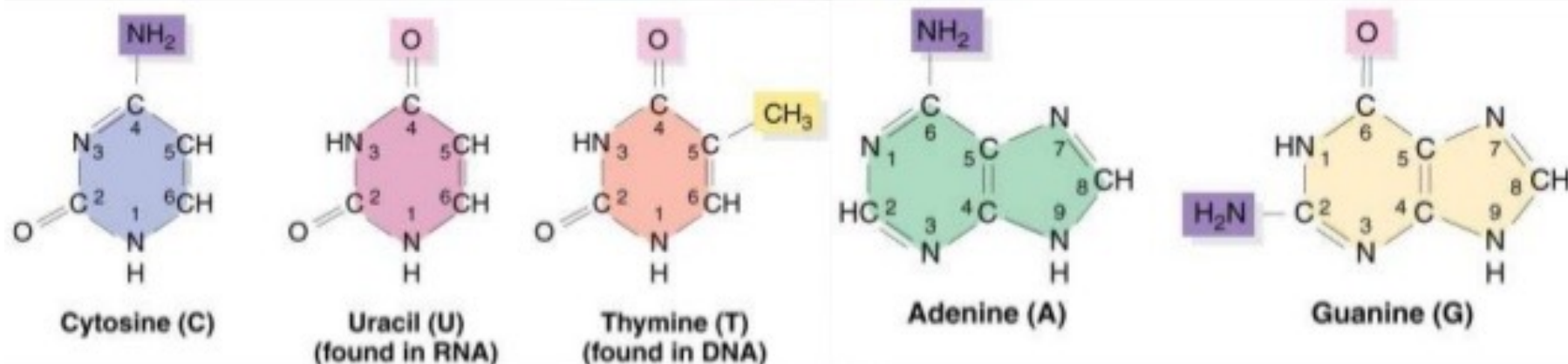
WHAT IS A NANOPORE?

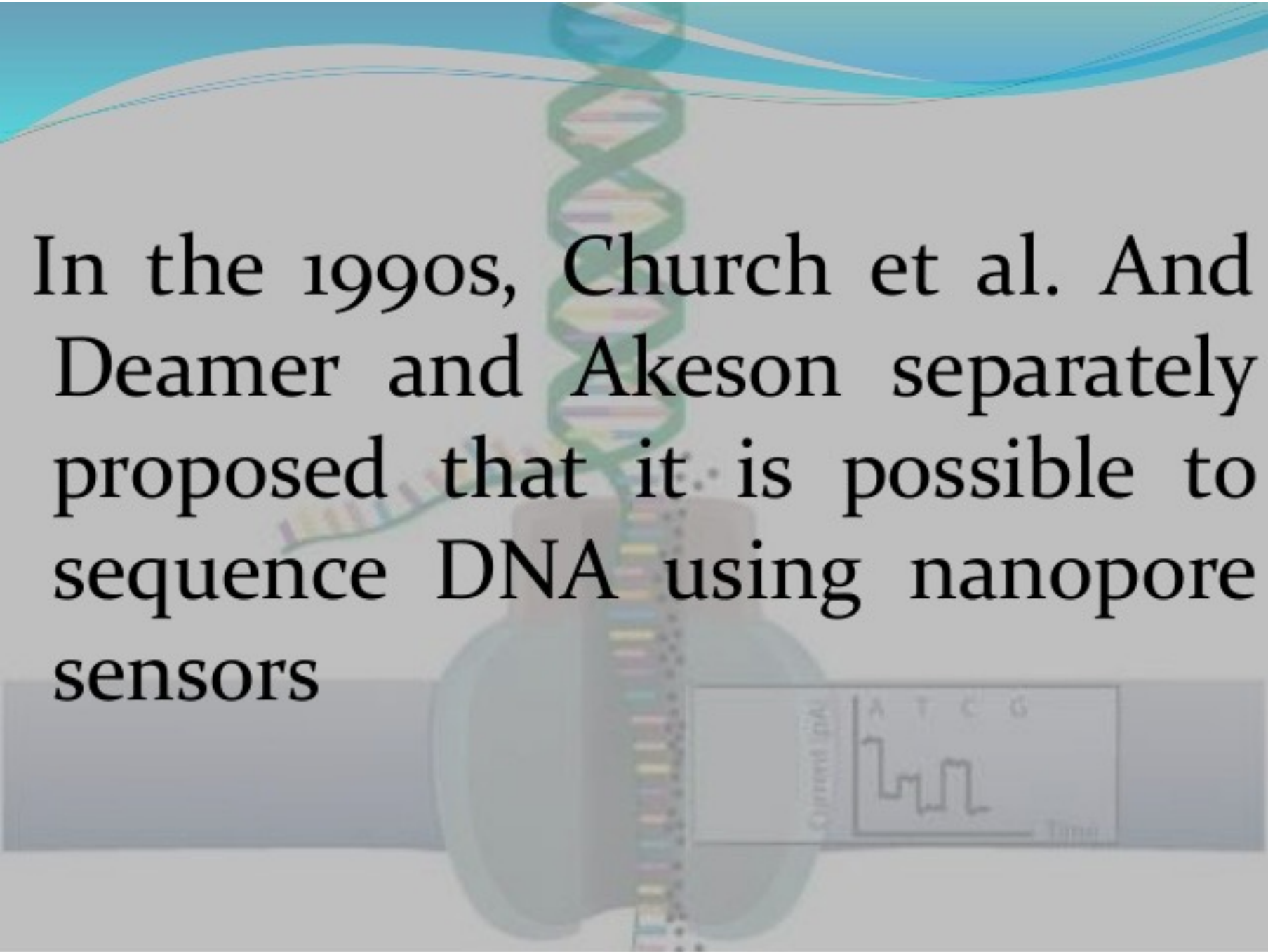
- Nanopore = ‘very small hole’
- Electrical current flows through the hole
- Introduce analyte of interest into the hole → identify “analyte” by the disruption or block to the electrical current



Conti...

- Since DNA bases (**adenine**, **thymine**, **cytosine** and **guanine**) are different from each other in atomic scale, it is essential to collect base-specific information at atomic level to correspond the DNA sequence with the measured signals. According to the different types of the signals, the detection methods can be roughly classified into two categories: the electrical detection methods and the optical readout method.





In the 1990s, Church et al. And Deamer and Akeson separately proposed that it is possible to sequence DNA using nanopore sensors

TYPES OF NANOPORES

Nanopore technologies can be broadly divided into two categories:

1. Biological nanopore
2. Solid-state nanopore

More recently, hybrid nanopores have been proposed to take advantage of the features of both biological and solid-state nanopores.

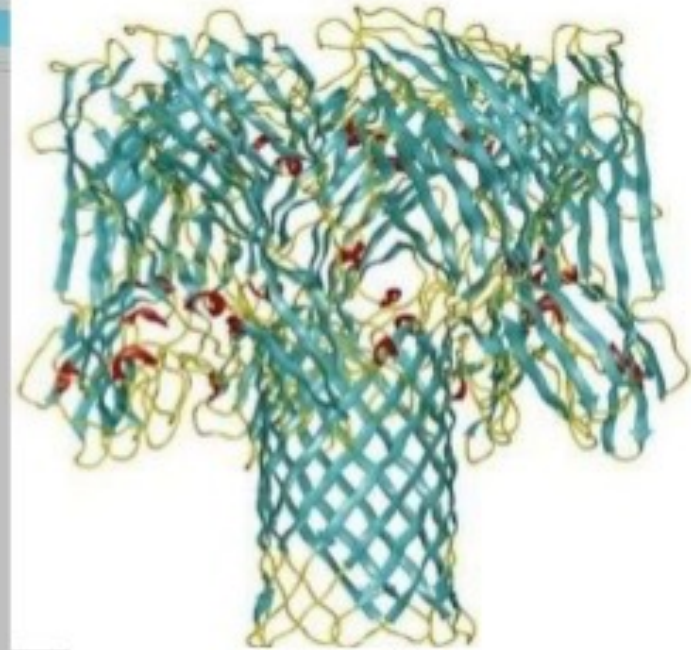
Biological nanopores

- also called transmembrane protein channels,
- are usually inserted into a substrate, such as planar lipid bilayers, liposomes, or other polymer films.
- Nanopores formed by pore-forming proteins
- Examples: alpha-hemolysin,
MspA,
phi 29

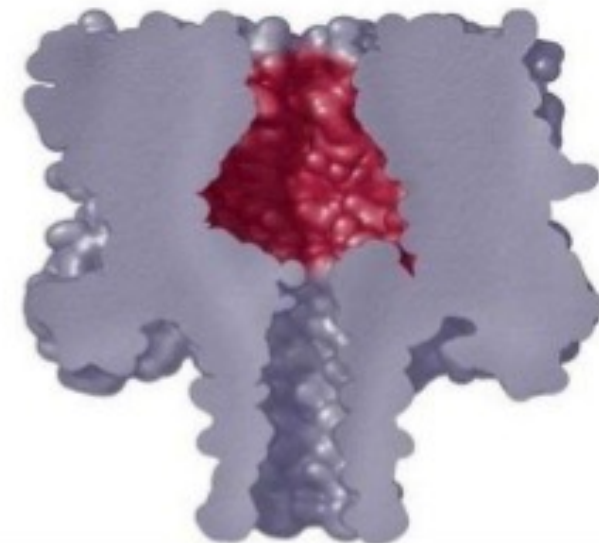
ALPHA-HEMOLYSIN

- α -Hemolysin (α -HL, also called α -toxin)
- α -HL is an exotoxin secreted by the bacterium *Staphylococcus aureus*, a human pathogen.
- This mushroom-shaped heptameric protein normally forms pores in the membranes.
- The limited pore size (1.4 nm) has restricted its application in the analysis of ssDNA, RNA, or small molecules.

A. Alpha hemolysin

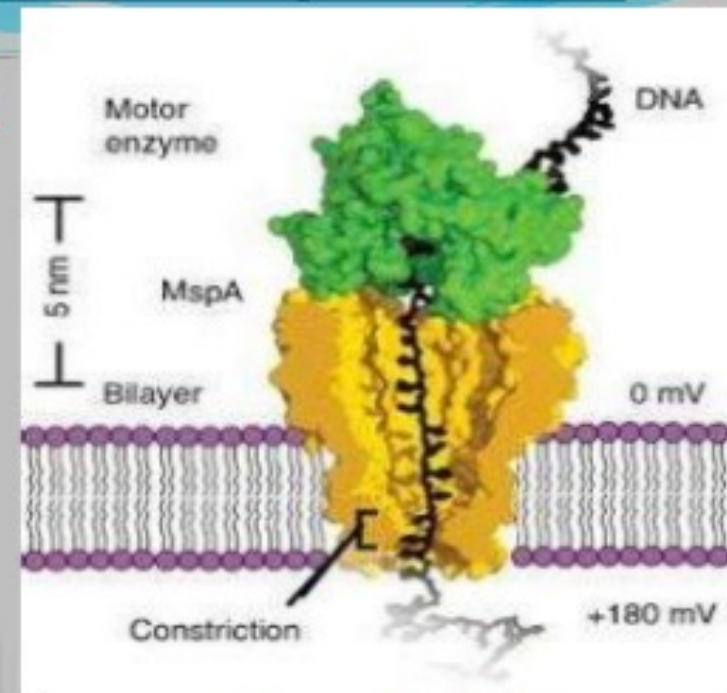


B. Cross section of alpha hemolysin

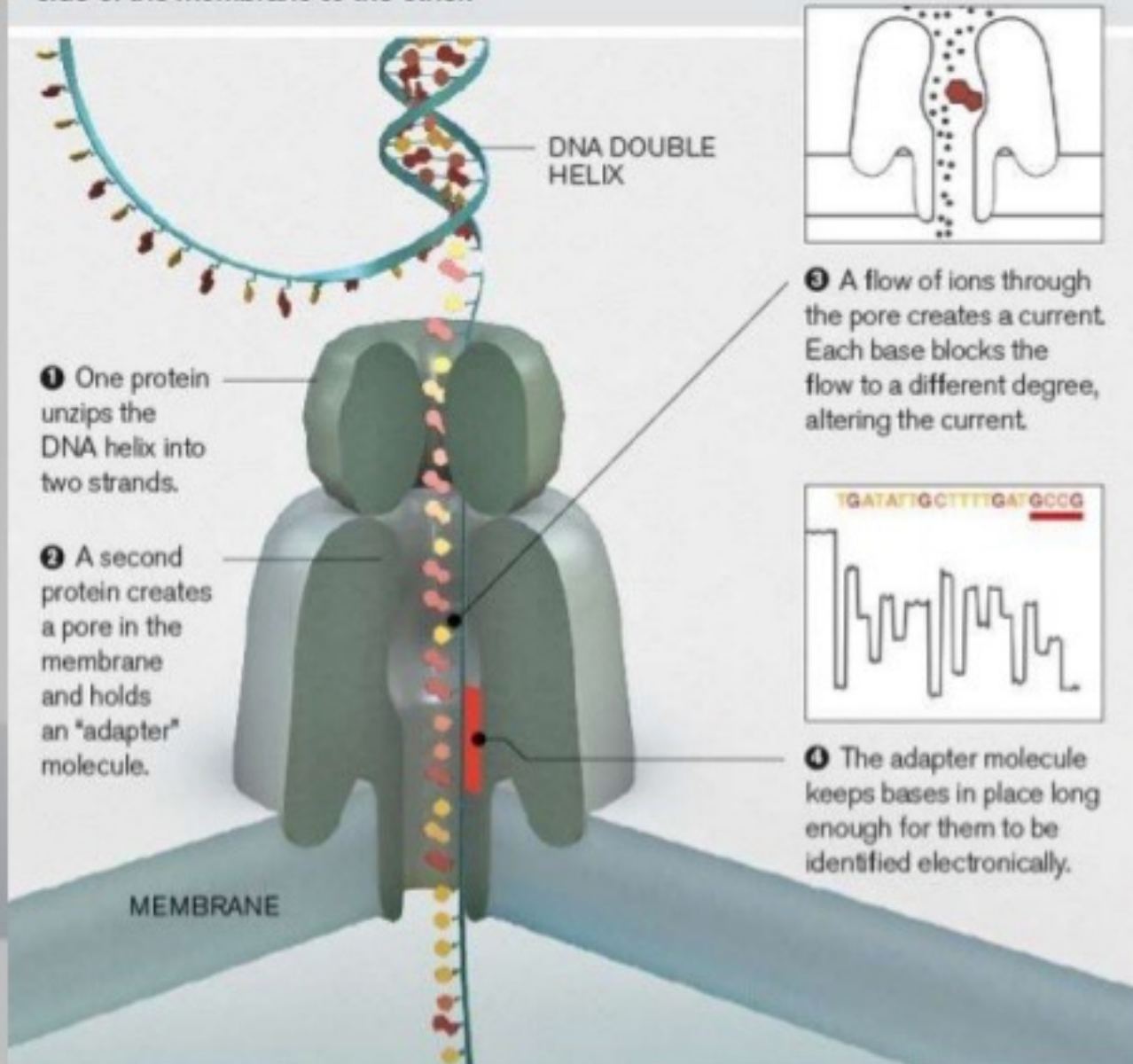


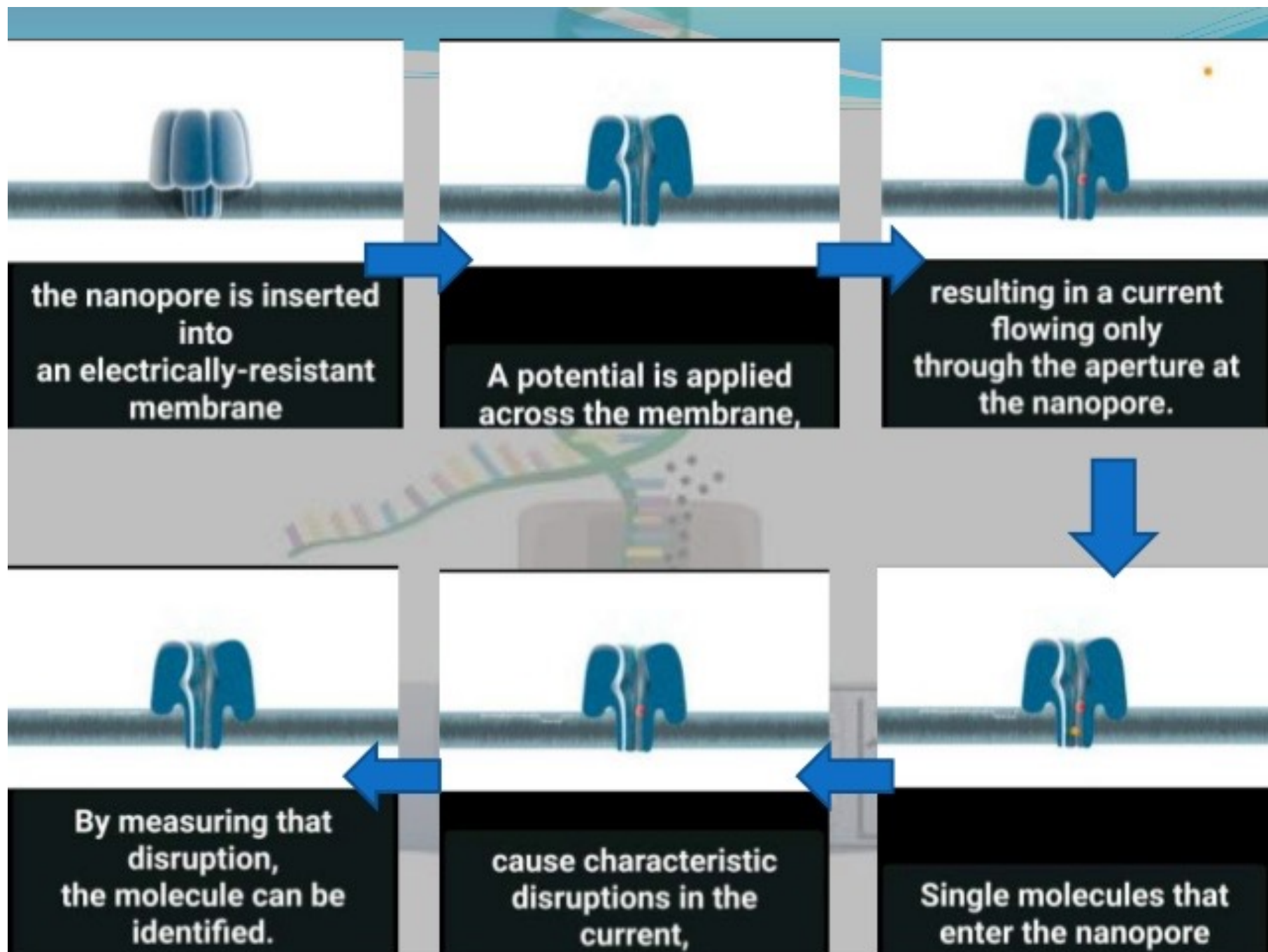
Mycobacterium smegmatis porin A (MspA)

- Powerful nanopore for reading information from four nucleotides simultaneously.
- by Butler et al. in 2008.
- The channel of the MspA octamer is 1 nm in diameter at the minimal point, which is relatively small and narrow, compared to that of alpha-HL. Thus, it can improve the spatial resolution of ssDNA sequencing.



DNA can be sequenced by threading it through a microscopic pore in a membrane. Bases are identified by the way they affect ions flowing through the pore from one side of the membrane to the other.





□ The advantages of biological nanopores include:

1. Their well-defined and highly-reproducible nanopore size and structure.
2. Biological nanopores can be modified easily with modern molecular biology techniques, such as mutating the nucleotide sequence to change the amino acid residue at a specific site.

• **Disadvantages of biological nanopores:**

1. The environmental demands(like temperature , electrolyte concentration and Ph) of nanopores to keep there biological activities.
2. The fragility of lipid bilayer makes the biological nanopore breakdown easily

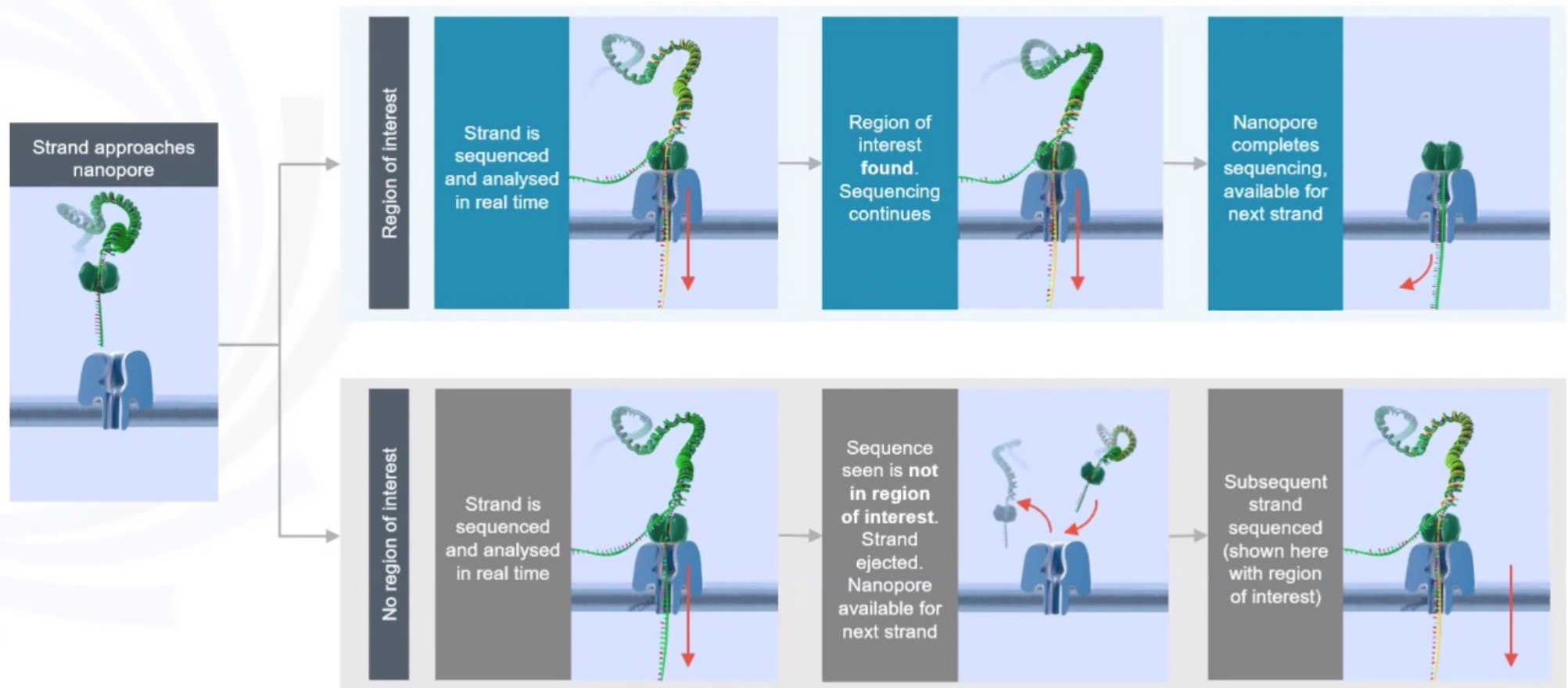
SOLID STATE NANOPORE

- In 2001, Li et al.
- are made in silicon compound membranes (Silicon nitride)
- More recently, the use of graphene as a material for solid-state nanopore sensing has been explored.

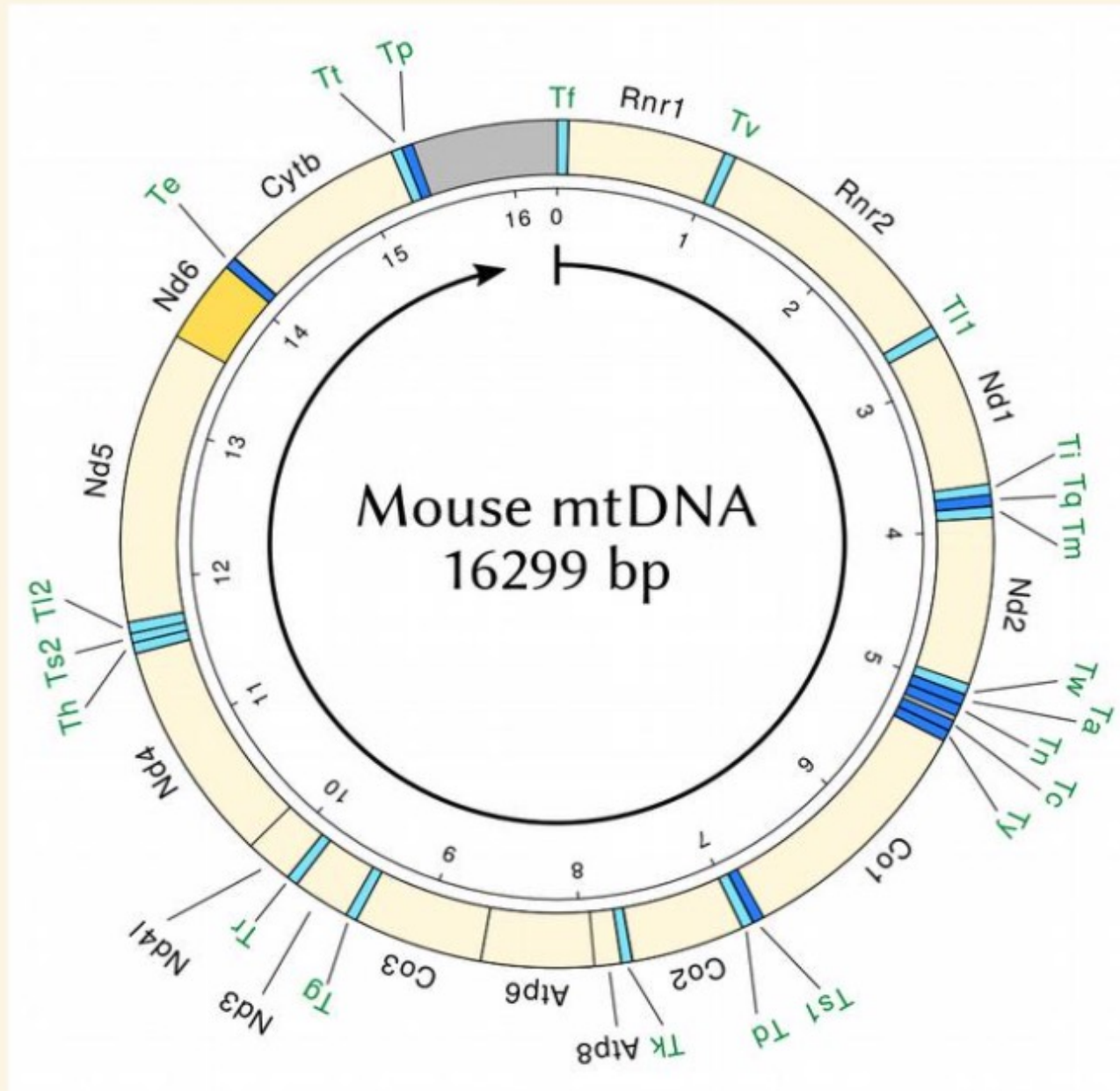


Method: Adaptive sampling

On-flow-cell enrichment, with no limit to read length or target number



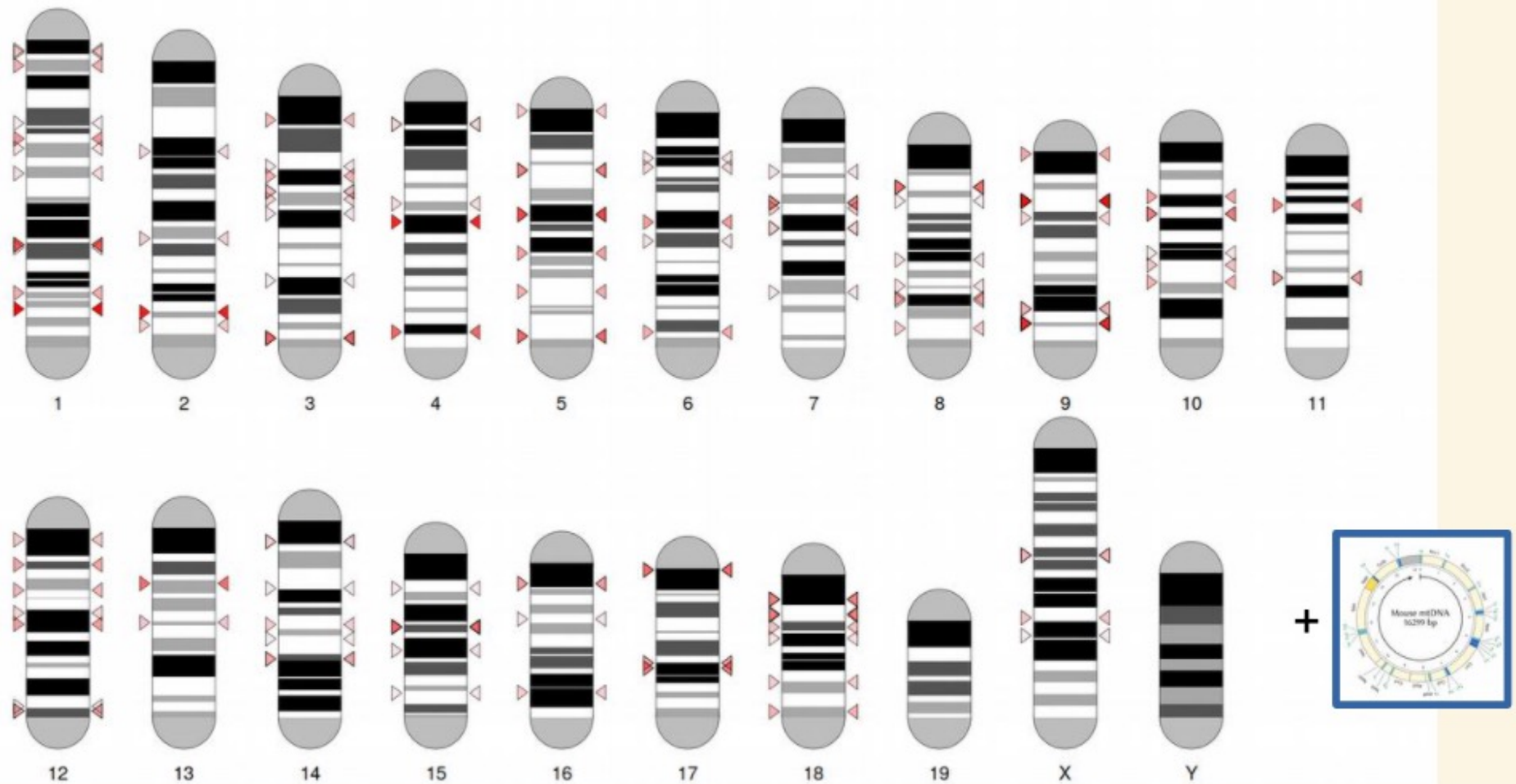
Mitochondrial Transcription



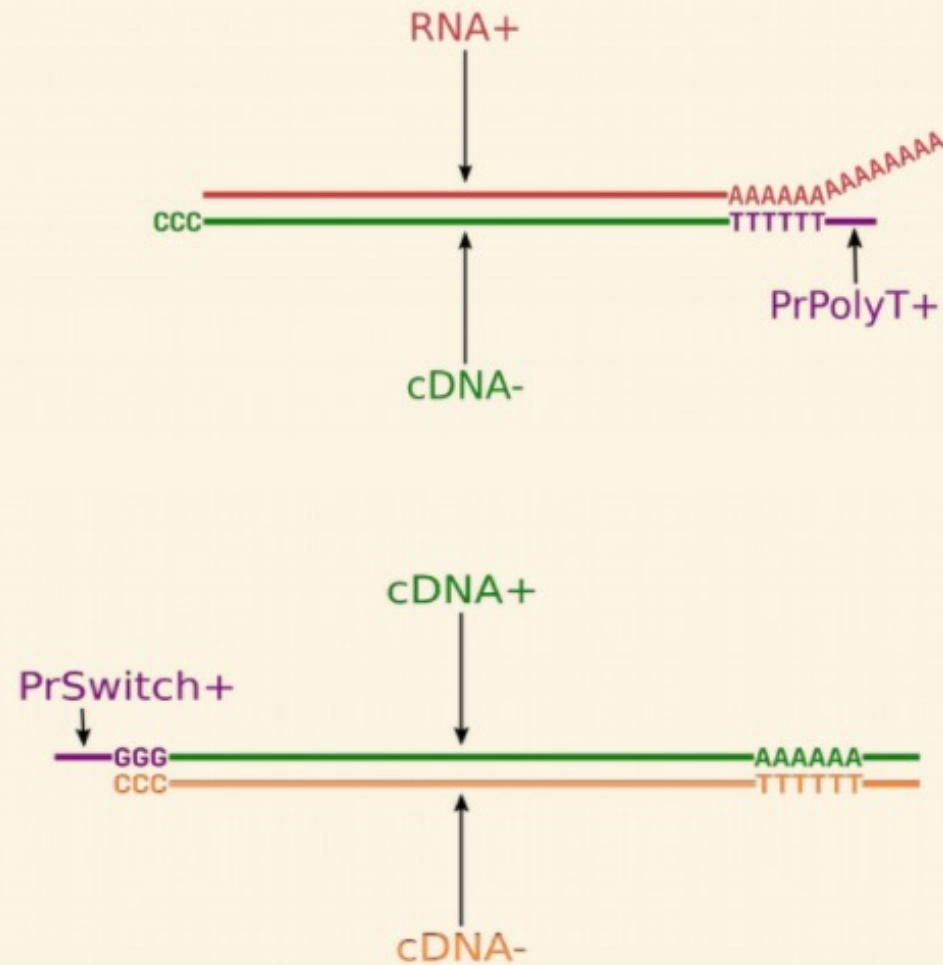
Mitochondrial Transcription



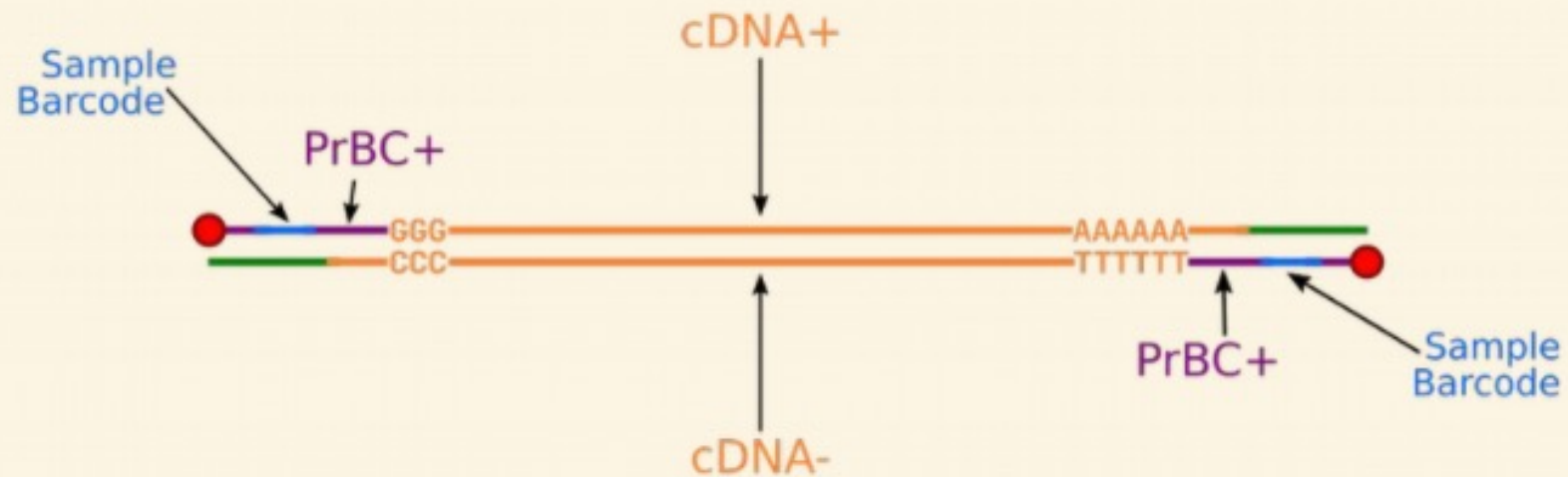
Mitochondrial duplication



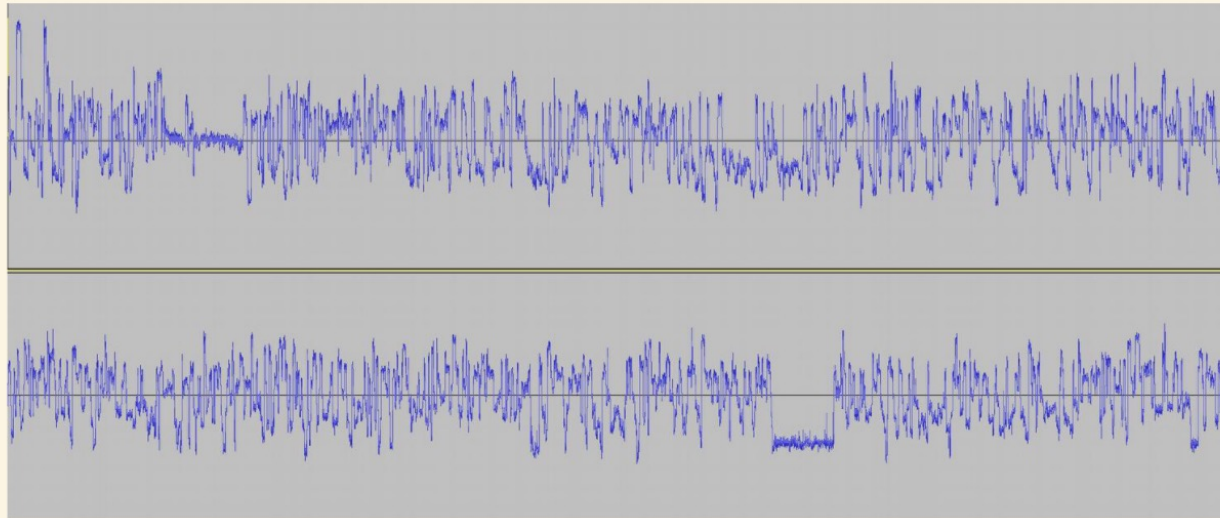
Strand-switching cDNA Generation



Barcoding & Amplification



Nanopore Raw Signal

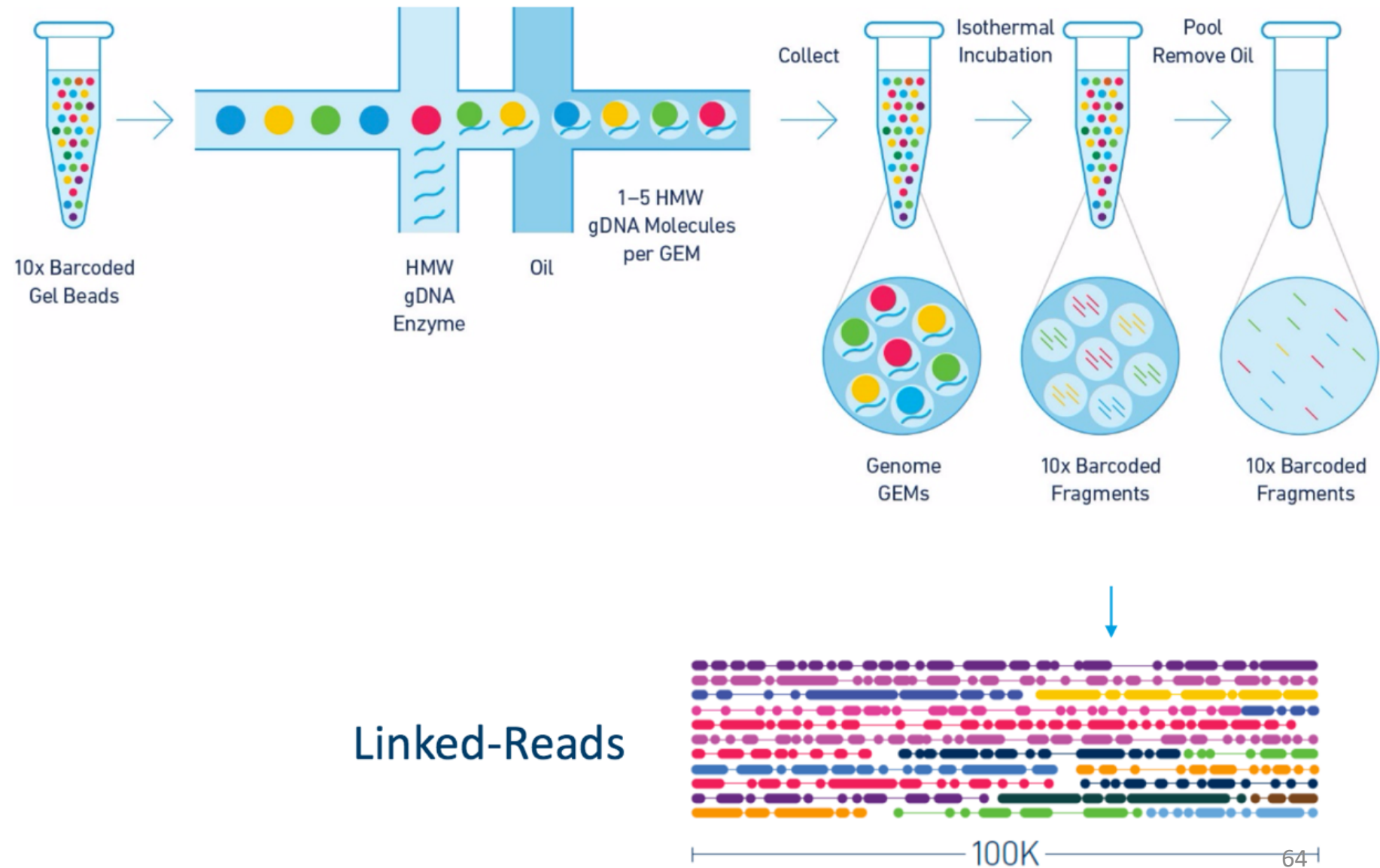


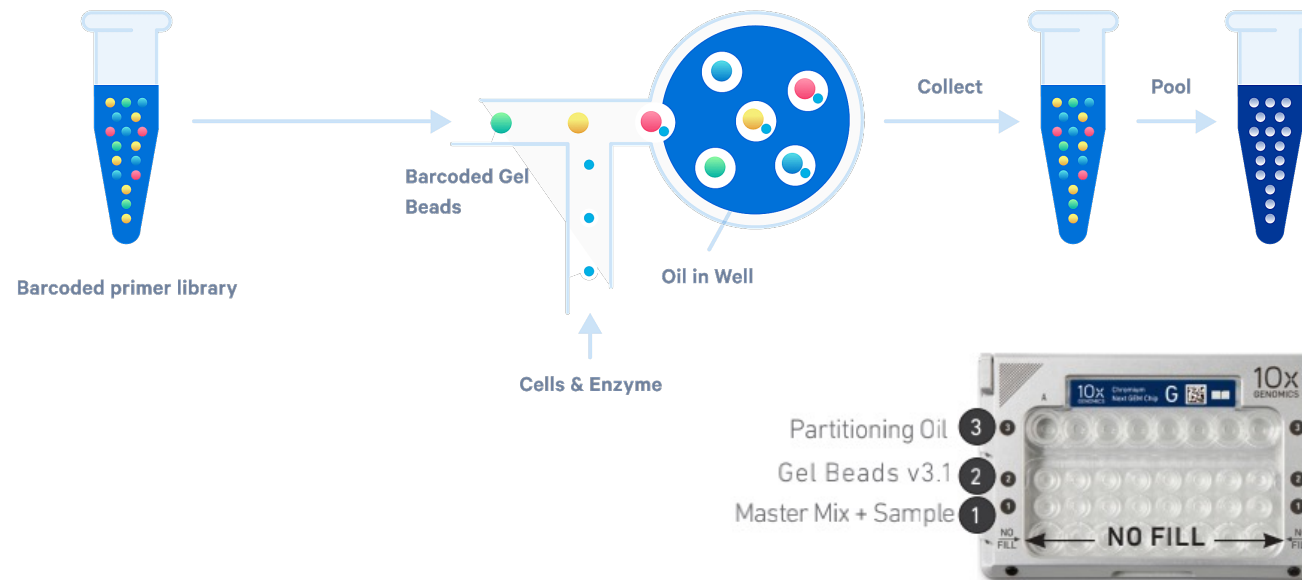
Nanopore Called Sequence

```
>3ec9fcb8-60cb-4620-bdba-3354c8ebe57e ch=303 read=26404 start_time=2017-12-21T10:14:26Z barcode=barcode07
CTCGTTTGATCGATTGCTTCGTTCCGGTCATGGAAGTGGTGTATTTATGGATCAATAGCCTACCGTGACGTGTACCGTGGAATGAATCATACTTGCCCTGTCGCTCATC
TTCTTTTTTTTTTTTTTTTTTTTTCTTTGTTTTTTCGGATTTGGTAATTTAATCAGTGATAACAGATGGTGTTCATGTTCCGTGTTCAGTGACCATAAACACACACA
TACACATATTCATCATGGTCTCCATACAACATATGTAAGAGGTAGACAATTCATTGTCCCCATTTTACAGACAAAAACCAAGGCTCAGAGAGAGTACCAGGCTTGTC
CCAGTCACACAGAGGAGGGAGGTAGAGGTTGAATGACAGGTAACTTTTCTTAGAGACTTGGGGACTGCTAGCTTCCCTTGGGGTACTAGAACCTCAGCCACCATTTTCT
GCATTTGTGAGCAGAGAACAGGAGTGGACAAGGGATGAAGGAAGAGGGGGGGTGGGCTGAGAGGAGGAGAGGGGCAGACGGTCATTTGAAACGAATGACTCTTCGGGA
TCCAGGGGTGTATTTGCAGATCTCTGAGTAGACAGATGGGTGCTGGGCAGCACCATGGGTAAATGCCCCACGAGGACGCCATGCAGGGGTGTATCACACACAGGGG
GGCCACCGGAGTCACCTGGCAAGAGTCTTGGTTTCCGTCTGCCTCAGCACAAATCATGCTGTTGGTGATCACACCTGGGTAGAAGGTCTCACACTGCTTCTGGCTAA
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AGGAGATCAGAAGTGGCTCTGCCCTGCAGTGTGGCTGCAAAAGTCTCTGACCCAGCAATATCCGGCACAGCAGAAAAAGGGATTCAATCCACCGGTAAACACG
TCACGGG
```

Chapter 4

Principle for 10x Genomics





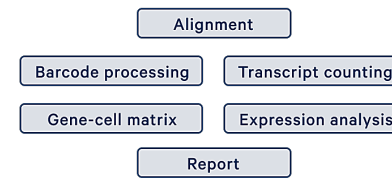
Cell or nuclei suspension



Barcoding & library construction



Sequence libraries



Cell Ranger pipelines



Report & visualization

Products



Single Cell Gene Expression
with CRISPR Screening & Cell Surface Protein



Single Cell Immune Profiling
with Antigen Specificity & Cell Surface Protein



Single Cell Multiome ATAC + Gene Expression



Single Cell ATAC



Single Cell CNV



Spatial Gene Expression
with Immunofluorescence



Spatial Proteomics



Targeted Gene Expression



Linked-Reads Genomics

PRODUCT NEWS

Product Updates

Instruments

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Cloud Analysis
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Multiomic Cytometry
Immune Receptor Mapping

Technology

Next GEM Technology
Visium Technology
In situ Technology

Compatible Products

Find 10x Compatible Products

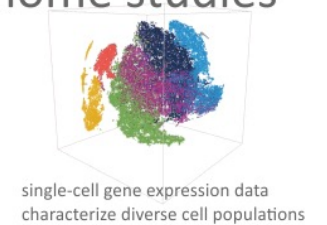
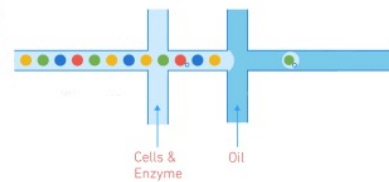
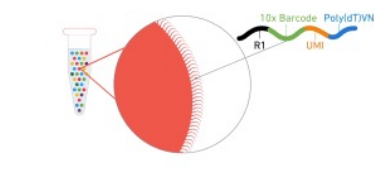
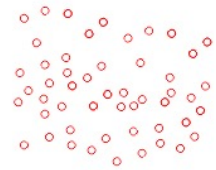
Certified Service Providers

Find a Service Provider
Become a Service Provider

Single-cell for 10x Genomics

10X Genomics nanodroplet technology for *single-cell* gene expression and genome studies

cell suspension

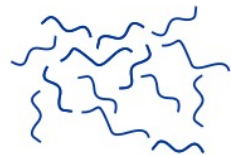


samples

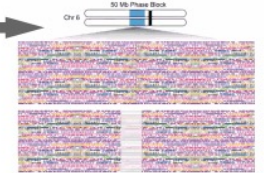
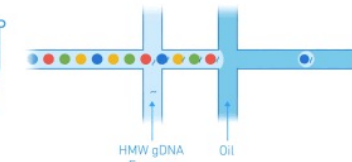
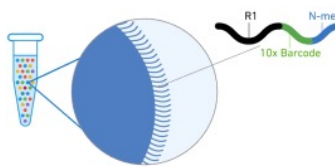
+ barcoded gel bead libraries

10X Genomics bead chip + Chromium Controller

Illumina compatible sequencing libraries



diluted HMW DNA



genomic linked-reads identify:
- structural genomic variations
- haplotype blocks
& enable high quality *de novo* genome assemblies

Single-cell RNA-seq

